

## Chapter 35

### JPEG Codec

ESP32-P4's JPEG codec is an image codec, which is based on the JPEG baseline standard, for compressing (encoding) and decompressing (decoding) images to reduce the bandwidth required to transmit images or the space required to store images, making it possible to process large-resolution images. The image encoding process mainly includes discrete cosine transform (DCT), zigzag scanning, quantization, and Huffman coding. The image decoding process, which is the inverse operation of the encoding process, mainly includes Huffman decoding, inverse quantization, inverse zigzag scanning, and inverse discrete cosine transform (IDCT).

#### 35.1 Terminology

|                         |                                                                                                                                                                                                                           |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MCU</b>              | Minimum coded unit. The basic unit of codec processing                                                                                                                                                                    |
| <b>baseline</b>         | A particular sequential DCT-based encoding and decoding process specified in the standard, which is required for all DCT-based decoding processes                                                                         |
| <b>image</b>            | A set of two-dimensional arrays of integer data                                                                                                                                                                           |
| <b>component</b>        | One of the two-dimensional arrays that comprise an image                                                                                                                                                                  |
| <b>data unit</b>        | An $8 \times 8$ block of pixels of one component                                                                                                                                                                          |
| <b>marker</b>           | A two-byte code in which the first byte is hexadecimal FF (0xFF) and the second byte is a value between 1 and hexadecimal FE (0xFE). It is served to identify the various structural parts of the compressed data formats |
| <b>restart marker</b>   | Used to isolate coded data segments and for application-specific uses, such as decoder re-synchronization and error recovery                                                                                              |
| <b>restart interval</b> | The integer number of MCUs processed between two restart markers                                                                                                                                                          |
| <b>scan</b>             | A single pass through the data for one or more of the components in an image                                                                                                                                              |

Figure 35.1-1 shows the schematic diagram of JPEG bitstream format. In this diagram, the dashed lines mark the structural composition of the corresponding parts of the bitstream, and the orange blocks indicate the marker codes.

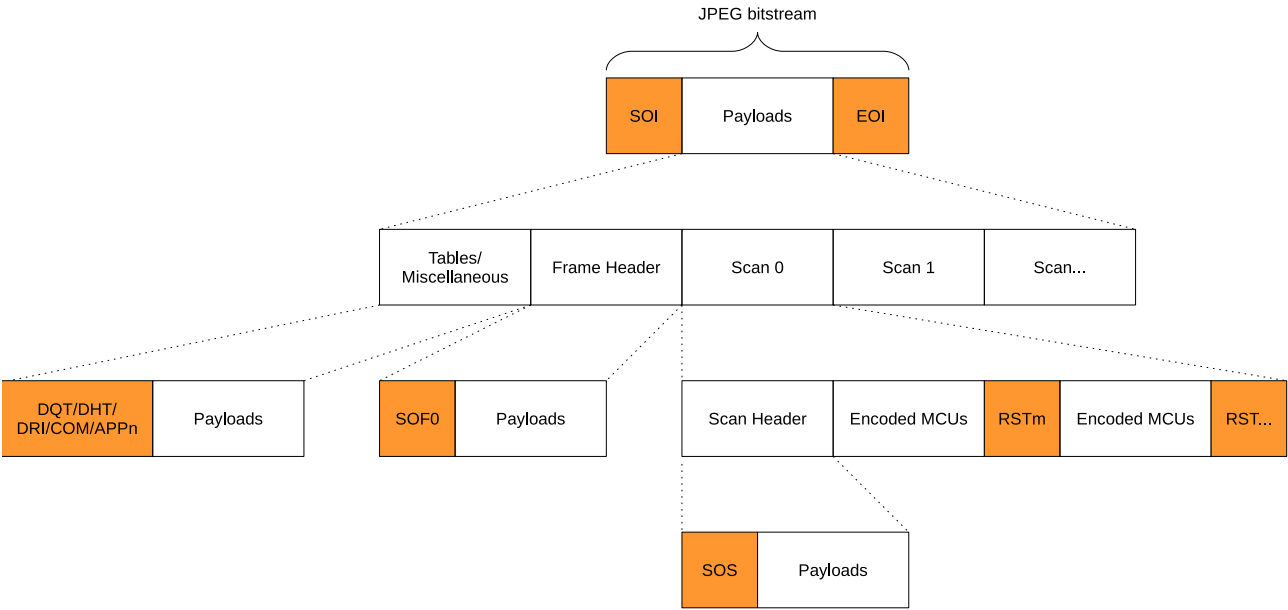


Figure 35.1-1. JPEG Bitstream Format

The marker code assignments are shown in Table 35.1-2.

Table 35.1-2. Marker Code Assignments

| Code Assignment | Marker | Description                                      |
|-----------------|--------|--------------------------------------------------|
| 0xFFC0          | SOF0   | Start of frame (Baseline DCT)                    |
| 0xFFC4          | DHT    | Define Huffman table(s)                          |
| 0xFFDm          | RSTm   | Restart with modulo 8 count “m”                  |
| 0xFFD8          | SOI    | Start of image                                   |
| 0xFFD9          | EOI    | End of image                                     |
| 0xFFDA          | SOS    | Start of scan                                    |
| 0xFFDB          | DQT    | Define quantization table(s)                     |
| 0xFFDD          | DRI    | Define restart interval                          |
| 0xFFEn          | APPn   | Reserved for application segments, “n” is 0 to F |
| 0xFFFE          | COM    | Comment                                          |

## 35.2 Introduction

The JPEG codec is based on the JPEG baseline standard, which specifies that an encoding process is completed through one or more scans. An scan contains one or more MCUs. MCU is the sequence of data units defined by the sampling factors of the image components in the scan, which means MCU of different image formats contains different sequence of data units, as shown in Table 35.2-1.

Table 35.2-1. The Number of Data Units in the MCU of Different Image Format

| Image Format | MCU <sup>1</sup>      |                         |                         |
|--------------|-----------------------|-------------------------|-------------------------|
|              | Luminance Component Y | Chrominance Component U | Chrominance Component V |
| YUV444       | 1 x 1                 | 1 x 1                   | 1 x 1                   |

|        |       |       |       |
|--------|-------|-------|-------|
| YUV422 | 2 x 1 | 1 x 1 | 1 x 1 |
| YUV420 | 2 x 2 | 1 x 1 | 1 x 1 |
| GRAY   | 1 x 1 | none  | none  |

<sup>1</sup> The MCU of each image format is represented by the number of data units in the horizontal direction x the number of data units in the vertical direction of each component.

At a time, the JPEG codec can only be configured as an encoder or a decoder, that is, it cannot work as an encoder and a decoder at the same time.

- When used as an encoder, it obtains the image to be compressed through 2D DMA and sends the encoded bitstream to the 2D DMA.
- When used as a decoder, it obtains the coded bitstream through 2D DMA and sends the decoded image to 2D DMA.

The workflow of JPEG codec is shown in Figure 35.2-1.

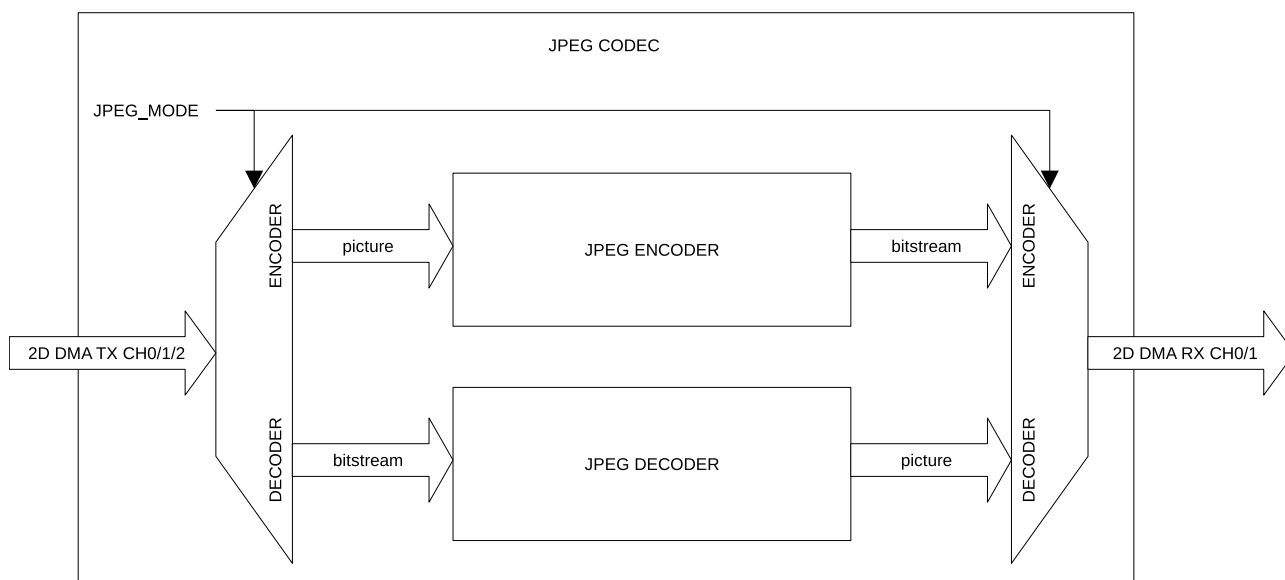


Figure 35.2-1. JPEG codec system connection block diagram

The function clock of the JPEG codec is SYS\_CLK.

## 35.3 Features

When used as an encoder, the JPEG codec has the following features:

- integrated discrete cosine transform algorithm
- integrated canonical Huffman coding
- RGB888, RGB565, YUV422 and GRAY as original input image formats
- conversion of RGB888 and RGB565 into YUV444, YUV422 or YUV420 (the only formats supported by impression) for image compression

- 4 x configurable quantization coefficient tables with 8-bit or 16-bit precision
- performance:
  - still image compression: up to 4K resolution
  - dynamic image compression: up to 1080P@40fps, 720P@70fps (excluding header encoding time)
- automatically added stuffed zero byte
- automatically added EOI marker

When used as a decoder, the JPEG codec has the following features:

- integrated inverse discrete cosine transform algorithm
- integrated Huffman decoding
- supported image formats for compressed bitstream decoding: YUV444, YUV422, YUV420, and GRAY.
- 4 x configurable quantization coefficient tables with 8-bit or 16-bit precision
- 2 x DC and 2 x AC Huffman tables
- supports image decoding of any resolution. However, the resolution of output decoded image differs from the format of the input image:
  - YUV444, GRAY: both the horizontal and vertical resolutions of the output decoded image are multiples of 8, i.e., 150 × 150 images with an output resolution of 152 × 152
  - YUV422: the horizontal resolution of the output decoded image is the multiples of 16 and the vertical resolution is multiples of 8, i.e., 150 × 150 images with an output resolution of 160 × 152
  - YUV420: both the horizontal and vertical resolutions of the output decoded image are multiples of 16, i.e., 150 × 150 images with an output resolution of 160 × 160
- performance:
  - still image decoding: up to 4K resolution
  - dynamic image decoding: up to 1080P@40fps, 720P@70fps (excluding header parsing time)

## 35.4 Architectural Overview

When the JPEG codec is configured as an encoder, its architecture is shown in Figure 35.4-1.

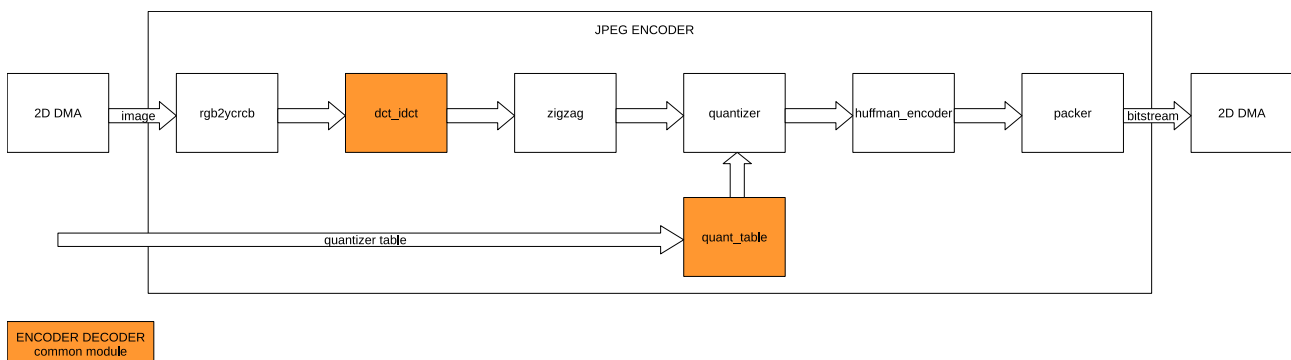


Figure 35.4-1. JPEG encoder architecture

- **rgb2ycrcb** implements the color space conversion, and converts the original image into an image format supported by the codec for compressing.
- **dct\_idct** implements two-dimensional DCT transformation of data units.
- **zigzag** performs zigzag scanning and sorting on the DCT transformed data units.
- **quant\_table** stores the quantization coefficient table delivered by the software.
- **quantizer** quantifies the zigzag sorted data units based on the quantization coefficient table stored in **quant\_table**.
- **huffman\_encoder** performs Huffman encoding on the quantized result of the data units.
- **packer** packs the Huffman-encoded data into groups of 32 bits, and outputs the final encoded bitstream.

When the JPEG codec is configured as a decoder, its architecture is shown in figure 35.4-2.

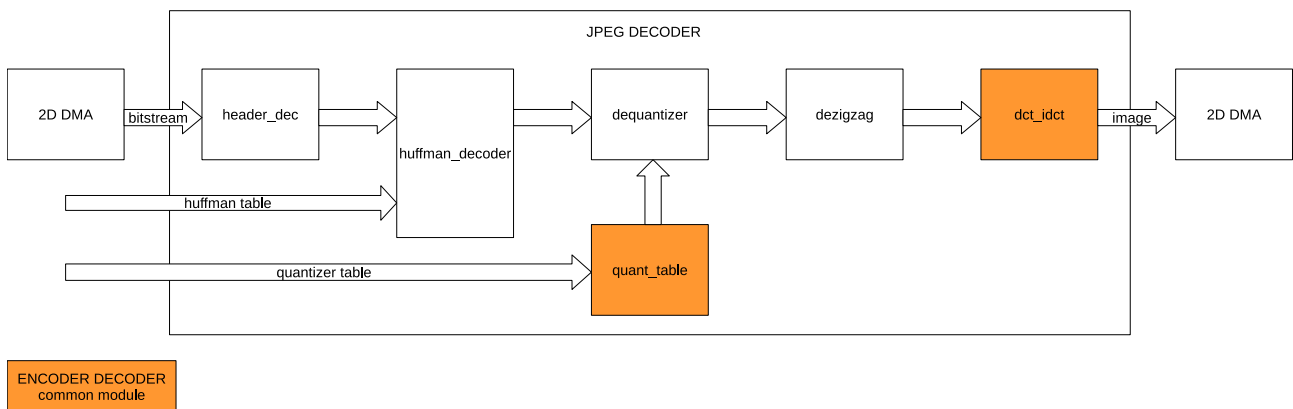


Figure 35.4-2. JPEG decoder architecture

- **header\_dec** implements bitstream extraction and marker SOS and EOI parsing, and controls the decoding process.
- **huffman\_decoder** stores the Huffman table delivered by the software, and implements Huffman decoding of the bitstream.
- **quant\_table** stores the quantization coefficient table delivered by the software.
- **dequantizer** dequantizes the Huffman decoded data based on the quantization coefficient table stored in **quant\_table**.
- **dezigzag** implements reverse zigzag scanning and sorting on the dequantized data units, and combines them into dequantized data units
- **dct\_idct** implements the two-dimensional inverse DCT transformation of the dequantized data units, and outputs the final decoded data units.

Since the JPEG codec cannot be used as an encoder and decoder at the same time, the encoder and decoder share the **quant\_table** and **dct\_idct** modules to save hardware resources.

## 35.5 Functional Description

The JPEG Codec can be configured to work as an encoder or a decoder.

## 35.5.1 JPEG Encoder

### 35.5.1.1 Pause

The JPEG encoder can be paused during the operation for other higher priority tasks. This feature is controlled by register field [JPEG\\_PAUSE\\_EN](#).

- Set [JPEG\\_PAUSE\\_EN](#) to 1, the JPEG encoder will no longer output the bitstream to the 2D DMA RX channel.
- Set [JPEG\\_PAUSE\\_EN](#) back to 0, the JPEG encoder will be resumed, and output the bitstream normally.

### 35.5.1.2 Color Space Conversion

According to the standard, the image formats supported by the JPEG encoder are YUV444, YUV422, YUV420, and GRAY. However, ESP32-P4's JPEG encoder has integrated a color space conversion function to convert RGB888 and RGB565 formats into the above-mentioned image formats supported by the encoder before encoding. In this way, the range of supported image formats is expanded.

The color space conversion formulas for converting RGB to YUV are:

$$\begin{aligned}Y &= (0.299 * R) + (0.587 * G) + (0.114 * B) \\U &= (-0.1687 * R) + (-0.3313 * G) + (0.5 * B) + 128 \\V &= (0.5 * R) + (-0.4187 * G) + (-0.0813 * B) + 128\end{aligned}$$

YUV formats only support downsampling. For example, YUV444 can remain unchanged or be downsampled to YUV422 or YUV420, and YUV422 can remain unchanged or be downsampled to YUV420.

The image format before conversion is represented by its format as well as pixel order. Table [35.5-1](#) demonstrates the representations of the supported image formats for color space conversion:

Table 35.5-1. Supported Image Formats for Color Space Conversion

| JPEG_EXTD_COLOR_SPACE_EN | JPEG_COLOR_SPACE | JPEG_EXTD_COLOR_SPACE | Format <sup>1</sup> | JPEG_PIXEL_REV | Pixel Order          |
|--------------------------|------------------|-----------------------|---------------------|----------------|----------------------|
| 0                        | 0                | N/A                   | RGB888              | 0              | RGB                  |
|                          |                  |                       |                     | 1              | BGR                  |
|                          | 1                |                       | YUV422 <sup>2</sup> | 0              | YUYV                 |
|                          |                  |                       |                     | 1              | UYVY                 |
|                          | 2                |                       | RGB565              | 0              | RGB                  |
|                          |                  |                       |                     | 1              | BGR                  |
|                          | 3                |                       | GRAY <sup>2</sup>   | N/A            | YYY                  |
| 1                        | N/A              | 0                     | YUV444              | 0              | YUV                  |
|                          |                  |                       |                     | 1              | VUY                  |
|                          |                  | 1                     | YUV420 <sup>2</sup> | N/A            | YYU/YYV <sup>3</sup> |
|                          |                  |                       |                     |                |                      |

<sup>1</sup> The image formats can be converted into other formats supported by the encoder via configuring register field [JPEG\\_SAMPLE\\_SEL](#):

- 0: YUV444
- 1: YUV422
- 2: YUV420

<sup>2</sup> Image format conversion is limited:

- YUV422: can only remain unchanged or be downsampled to YUV420. Setting [JPEG\\_SAMPLE\\_SEL](#) to 0 has no effect.
- YUV420 or GRAY: will not be converted, regardless of the value of [JPEG\\_SAMPLE\\_SEL](#).

<sup>3</sup> The pixel arrangement alternates between rows: odd-numbered rows use the YYU pattern, and even-numbered rows use the YYV pattern.

### 35.5.1.3 Configurable Quantization Coefficient Table

ESP32-P4 JPEG encoder has implemented four 8 x 8 quantization coefficient tables in its hardware, which allows 64 software-configurable coefficient values per table.

The precision of each quantization table is configurable via [JPEG\\_QNR\\_PRECISION](#):

- 0: low 8-bit effective
- 1: low 16-bit effective

The coefficient values for each table can be configured in FIFO mode or non-FIFO mode depending on the value of [JPEG\\_QNR\\_FIFO\\_EN](#):

- 0: non-FIFO mode, in which each coefficient is written to a specific address
- 1: FIFO mode, in which all coefficients are written to the same address

Table 35.5-2. Position Number of Coefficients in a Quantization Coefficient Table (Example)

|     |   |    |    |    |    |    |    |
|-----|---|----|----|----|----|----|----|
| 0   | 1 | 2  | 3  | 4  | 5  | 6  | 7  |
| 8   | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| ... |   |    |    |    |    |    |    |

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 |
|----|----|----|----|----|----|----|----|

In different modes, the configuration of the quantization coefficient table is different. Note,  $n$  ( $n = 0, 1, 2, 3$ ) indicates the number of each quantization coefficient table:

- non-FIFO mode: each quantization coefficient in a specific position of a quantization coefficient table  $n$  must be written to a specific address, which is (JPEG codec base address +  $n * 0x100 + \text{quantization coefficient position number} * 0x4$ ), in which the *quantization coefficient position number* can be 0, 1, 2, ..., 63 (see Table 35.5-2). In this mode, since each coefficient is written to a specific address, configuring the 64 quantization coefficients in sequence is not required.
- FIFO mode: all 64 quantization coefficients of the quantization table  $n$  are written to the same address, which is `JPEG_TnQNR_REG`. In this mode, since all coefficients are written to the same address, configuring the 64 quantization coefficients one by one in sequence is required. Afterwards, the hardware will automatically store these 64 quantization coefficients in sequence as shown in Table 35.5-2.

If there are less than 4 quantization tables, the corresponding quantization table will not be written.

To use a specific quantization coefficient table  $n$  for the luminance component or the chrominance component, just set the corresponding register fields to  $n$ :

- luminance component: `JPEG_LQNR_TBL_SEL`
- chrominance component: `JPEG_CQNR_TBL_SEL`

The JPEG baseline standard recommended 8-bit precision quantization coefficient tables for the luminance component and the chrominance component are shown in Table 35.5-3 and Table 35.5-4.

**Table 35.5-3. Recommended 8-bit Precision Quantization Coefficient Table for the Luminance Component**

|    |    |    |    |     |     |     |     |
|----|----|----|----|-----|-----|-----|-----|
| 16 | 11 | 10 | 16 | 24  | 40  | 51  | 61  |
| 12 | 12 | 14 | 19 | 26  | 58  | 60  | 55  |
| 14 | 13 | 16 | 24 | 40  | 57  | 69  | 56  |
| 14 | 17 | 22 | 29 | 51  | 87  | 80  | 62  |
| 18 | 22 | 37 | 56 | 68  | 109 | 103 | 77  |
| 24 | 35 | 55 | 64 | 81  | 104 | 113 | 92  |
| 49 | 64 | 78 | 87 | 103 | 121 | 120 | 101 |
| 72 | 92 | 95 | 98 | 112 | 100 | 103 | 99  |

**Table 35.5-4. Recommended 8-bit Precision Quantization Coefficient Table for the Chrominance Component**

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| 17 | 18 | 24 | 47 | 99 | 99 | 99 | 99 |
| 18 | 21 | 26 | 66 | 99 | 99 | 99 | 99 |
| 24 | 26 | 56 | 99 | 99 | 99 | 99 | 99 |
| 47 | 66 | 99 | 99 | 99 | 99 | 99 | 99 |
| 99 | 99 | 99 | 99 | 99 | 99 | 99 | 99 |
| 99 | 99 | 99 | 99 | 99 | 99 | 99 | 99 |
| 99 | 99 | 99 | 99 | 99 | 99 | 99 | 99 |



|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| 99 | 99 | 99 | 99 | 99 | 99 | 99 | 99 |
|----|----|----|----|----|----|----|----|

More information can be found in [ITU T.81 Recommendation](#). Adjusting the quantization table can change the compression ratio (the ratio between the uncompressed image size and compressed image size) and the quality of encoded images (the larger the quantization coefficient, the greater the compression ratio and the worse the image quality).

#### 35.5.1.4 Stuffed Zero Byte

As required in the JPEG baseline standard to avoid any marker within a Huffman-coded segment, any 0xFF byte generated by the Huffman coding should be followed by a “stuffed” zero byte.

JPEG encoder has implemented a mechanism to automatically add this “stuffed” 0x00 byte after each 0xFF byte in the Huffman-coded segment of bitstream.

Users can configure whether to add 0x00 byte after each 0xFF byte via [JPEG\\_FF\\_CHECK\\_EN](#):

- 0: not add
- 1: add (by default)

#### 35.5.1.5 EOI Marker

The end of Image (EOI) marker is used to mark the end of the bitstream of a compressed image. The EOI marker code is 0xFFD9.

JPEG encoder has implemented a mechanism to automatically add the EOI marker at the end of the bitstream.

Users can configure whether to add EOF marker via [JPEG\\_TAILER\\_EN](#):

- 0: not add
- 1: add

### 35.5.2 JPEG Decoder

The information in the bitstream must be parsed by the software, then sent to the hardware through the APB configuration bus. Note that the hardware does not support parsing any marker segments, except the SOS, RST and EOI marker segments.

#### 35.5.2.1 Multiple Chrominance Components

The frame header contains information about the number of chrominance components of the image, the ID of each component, and the sampling factors in the horizontal and vertical directions of each component.

JPEG decoder's software must parse such information and pass it to respective registers [JPEG\\_COMPONENT\\_NUM](#), [JPEG\\_Cn\\_ID](#), [JPEG\\_Cn\\_X\\_FACTOR](#), and [JPEG\\_Cn\\_Y\\_FACTOR](#). Here, *n* can be 0, 1, 2, indicating that up to 3 chrominance components are supported by hardware. Note that if less than 3 components are involved, just do not configure the rest of the registers.

According to the value of register [JPEG\\_Cn\\_ID](#) of each chrominance component, the hardware will check whether it is consistent with the ID in the scan header information. The sampling factor indicates the number

of data units of this component that an MCU contains in both the horizontal and vertical directions. So according to the value of register [JPEG\\_COMPONENT\\_NUM](#), [JPEG\\_Cn\\_X\\_FACTOR](#) and [JPEG\\_Cn\\_Y\\_FACTOR](#), the hardware can obtain the proportional relationship of chrominance components, that is, the format of the image to be decoded. See Table 35.5-5.

Table 35.5-5. Sampling Factor and Image Format

| JPEG_COMPONENT_NUM | JPEG_Cn_X/Y_FACTOR                                                                                                   | Format of the Image to be Decoded |
|--------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| 0                  | -                                                                                                                    | invalid                           |
| 1                  | CO_X_FACTOR is 1<br>CO_Y_FACTOR is 1                                                                                 | GRAY                              |
| 2                  | -                                                                                                                    | invalid                           |
| 3                  | CO_X_FACTOR is 1<br>CO_Y_FACTOR is 1<br>C1_X_FACTOR is 1<br>C1_Y_FACTOR is 1<br>C2_X_FACTOR is 1<br>C2_Y_FACTOR is 1 | YUV444                            |
|                    | CO_X_FACTOR is 2<br>CO_Y_FACTOR is 1<br>C1_X_FACTOR is 1<br>C1_Y_FACTOR is 1<br>C2_X_FACTOR is 1<br>C2_Y_FACTOR is 1 | YUV422                            |
|                    | CO_X_FACTOR is 2<br>CO_Y_FACTOR is 2<br>C1_X_FACTOR is 1<br>C1_Y_FACTOR is 1<br>C2_X_FACTOR is 1<br>C2_Y_FACTOR is 1 | YUV420                            |

Regardless of the format of the image to be decoded, the decoder only supports one scan, which contains all components. After decoding, the image will be sent to 2D DMA, so please refer to Chapter 6 [2D-DMA Controller \(2D-DMA\)](#) for the layout of the decoded image in each format.

### 35.5.2.2 Parsing RST Marker

The bitstream may have a restart (RST) marker in the scan segment. The RST marker code is 0xFFD $x$  ( $x = 0, 1, \dots, 7$ ).

JPEG decoder's hardware can parse RST markers and check whether the actual number of MCUs between two RST markers is the same as the restart interval defined in the DRI segment of the Tables/Miscellaneous segment.

Firstly, the software parses the DRI segment and pass the defined restart interval to register [JPEG\\_RESTART\\_INTERVAL](#). Then, the hardware will check whether the subsequent bitstream has an RST marker every time [JPEG\\_RESTART\\_INTERVAL](#) MCUs are decoded.

Configuring `JPEG_RESTART_INTERVAL` to 0 indicates the bitstream does not contain any RST marker. In this case, the hardware will not check RST markers.

### 35.5.2.3 Configurable Quantization Coefficient Table

ESP32-P4 JPEG decoder also supports 4 configurable quantization coefficient tables. Since the JPEG codec cannot be used as an encoder and decoder at the same time, decoder can share the same four configurable quantization coefficient tables with the encoder to save hardware resources.

Therefore, the configuration of decoder's quantization coefficient tables is similar to that of the encoder (see Section 35.5.1.3) with some exceptions described below:

- The software needs to parse the DQT segment in the Tables/Miscellaneous segment to obtain the quantization coefficient tables of the decoder. The recommended quantization coefficient tables are the same as the encoder shown in Table 35.5-3 and Table 35.5-4.
- The precision of the quantization table $n$  is configured via higher 4 bits of register `JPEG_T $n$ _DQT_INFO`:
  - 0: low 8-bit effective
  - 1: low 16-bit effective

The precision is also obtained from the DQT segment in the Tables/Miscellaneous segment.

- For each chrominance component, a different quantization coefficient table can be selected. The lower 4 bits of register `JPEG_T $n$ _DQT_INFO` specify the ID of each quantization coefficient table, and each chrominance component selects the quantization coefficient table whose ID is the value of register `JPEG_C $n$ _DQT_TBL_SEL`. The ID of each quantization coefficient table is also obtained from the DQT segment in the Tables/Miscellaneous segment.

### 35.5.2.4 Configurable Huffman Table

ESP32-P4 supports up to 2 DC Huffman tables and 2 AC Huffman tables. Users can configure as many Huffman tables as needed within the limits.

Each Huffman table can be established with three types of information, which can be obtained from the DHT segment in the Tables/Miscellaneous segment:

- the number of codeword with a codeword length of 1 to 16 bits,
- the symbols corresponding to the decoded codeword
  - DC table: up to 16 symbols
  - AC table: up to 256 symbols
- the minimum codeword with a codeword length of 1 to 16 bits.

Both of the DC and AC Huffman tables can be configured in a FIFO fashion or a non-FIFO fashion depending on the value of `JPEG_DHT_FIFO_EN`:

- 0: non-FIFO mode
- 1: FIFO mode

In different modes, the configuration of the Huffman table is different. Here is the configuration of DC tables $n$  ( $n = 0, 1$ ):

- non-FIFO mode: each piece in different type of information used to establish the DC $n$  Huffman table must be written to a specific address.
  - the number of codeword with a codeword length of 1 to 16 bits: JPEG codec base address + 0x500 +  $n * 0x80 + (\text{codeword length} - 1) * 0x4$ , in which *codeword length* = 1, 2, ..., 16.
  - the symbols corresponding to the decoded codeword: JPEG codec base address + 0x600 +  $n * 0x40 + \text{symbol number} * 0x4$ , in which *symbol number* = 0, 1, ..., 15.
  - the minimum codeword with a codeword length of 1 to 16 bits: JPEG codec base address + 0xF00 +  $n * 0x80 + (\text{codeword length} - 1) * 0x4$ , in which *codeword length* = 1, 2, ..., 16.

In this mode, since each piece of different type of information is written to a specific address, configuring each piece of information in sequence is not required.

- FIFO mode: all pieces of the same type of information used to establish the DC $n$  Huffman table must be written to the same address, respectively.
  - the number of codeword with a codeword length of 1 to 16 bits: [JPEG\\_DHT\\_TOTLEN\\_DC \$n\$ \\_REG](#).
  - all symbols corresponding to the decoded codeword (DC table has up to 16 symbols): [JPEG\\_DHT\\_VAL\\_DC \$n\$ \\_REG](#).
  - the minimum codeword with a codeword length of 1 to 16 bits: [JPEG\\_DHT\\_CODEMIN\\_DC \$n\$ \\_REG](#).

In this mode, since all pieces of the same type of information are written to the same address, configuring each piece of same type of information one by one in sequence is required. Afterwards, the hardware will automatically store such information in sequence.

The configuration of AC $n$  ( $n = 0, 1$ ) tables is very similar to that of the DC $n$  ( $n = 0, 1$ ) tables. Here below is the configuration of AC $n$  ( $n = 0, 1$ ) tables:

- non-FIFO mode: each piece in different type of information used to establish the AC $n$  Huffman table must be written to a specific address.
  - the number of codeword with a codeword length of 1 to 16 bits: JPEG codec base address + 0x540 +  $n * 0x80 + (\text{codeword length} - 1) * 0x4$ , in which *codeword length* = 1, 2, ..., 16.
  - the symbols corresponding to the decoded codeword: JPEG codec base address + 0x680 +  $n * 0x400 + \text{symbol number} * 0x4$ , in which *symbol number* = 0, 1, ..., 256.
  - the minimum codeword with a codeword length of 1 to 16 bits: JPEG codec base address + 0xF40 +  $n * 0x80 + (\text{codeword length} - 1) * 0x4$ , in which *codeword length* = 1, 2, ..., 16.

In this mode, since each piece of different type of information is written to a specific address, configuring each piece of information in sequence is not required.

- FIFO mode: all pieces of the same type of information used to establish the AC $n$  Huffman table must be written to the same address, respectively.
  - the number of codeword with a codeword length of 1 to 16 bits: [JPEG\\_DHT\\_TOTLEN\\_AC \$n\$ \\_REG](#).
  - all symbols corresponding to the decoded codeword (AC table has up to 256 symbols): [JPEG\\_DHT\\_VAL\\_AC \$n\$ \\_REG](#).
  - the minimum codeword with a codeword length of 1 to 16 bits: [JPEG\\_DHT\\_CODEMIN\\_AC \$n\$ \\_REG](#).

In this mode, since all pieces of the same type of information are written to the same address, configuring each piece of same type of information one by one in sequence is required. Afterwards, the hardware will automatically store such information in sequence.

For the minimum codeword with a codeword length of 1 to 16 bits, it should be moved left to the high bit of the 16-bit configuration data. If a codeword length does not exist, the configuration value is 0xFFFF. For example, if a minimum codeword is 0x7C, the 16-bit configuration data should be 0x7C00.

The JPEG baseline standard recommended DC and AC Huffman tables can be found in [ITU T.81 Recommendation](#).

The software can obtain the IDs of DC $n$  and AC $n$  Huffman tables by parsing the DHT segment in the Tables/Miscellaneous segment, then pass these ID to the corresponding register `JPEG_DC $n$ _DHT_ID` and `JPEG_AC $n$ _DHT_ID`.

The hardware will automatically use the DC $n$  and AC $n$  Huffman tables with the same ID from `JPEG_DC $n$ _DHT_ID` and `JPEG_AC $n$ _DHT_ID`.

### 35.5.2.5 Timeout Detection

Additionally, timeout detection is supported when the JPEG codec is working as a decoder. The hardware triggers the timeout interrupt `JPEG_DECODE_TIMEOUT_INT`:

- when the decoder stops reading the bitstream for more than  $(2^{JPEG\_DECODE\_TIMEOUT\_THRES} - 1)$  clock cycles,
- or when the number of clock cycles consumed for decoding an image exceeds the value of  $(2^{JPEG\_DECODE\_TIMEOUT\_THRES} - 1)$

In this case, the decoder will be reset depending on the configuration of `JPEG_DECODE_TIMEOUT_TASK_SEL`:

- 0: the hardware will not perform any operation, and the software is required to configure the `JPEG_SOFT_RST` or `JPEG_FSM_RST` to perform a reset operation.
- 1: the hardware will automatically reset the decoder to exit the timeout state and enter the initial state.

## 35.6 Interrupts

ESP32-P4's JPEG Codec can generate the `JPEG_INTR` interrupt signal that will be sent to the [Interrupt Matrix](#).

Interrupt signal `JPEG_INTR` can be generated by the following internal interrupt sources:

- `JPEG_DONE_INT`: Triggered when a frame of image encoding or decoding ends.
- `JPEG_RLE_PARALLEL_ERR_INT`: Triggered when a run-length encoding error occurs.
- `JPEG_CID_ERR_INT`: Triggered when the decoded component ID is different from the component ID configured by the software.
- `JPEG_C_DHT_DC_ID_ERR_INT`: Triggered when the decoded DC Huffman table ID of each component is not the software configured DCO Huffman table ID or DC1 Huffman table ID.

- `JPEG_C_DHT_AC_ID_ERR_INT`: Triggered when the decoded AC Huffman table ID of each component is not the software configured ACO Huffman table ID or AC1 Huffman table ID.
- `JPEG_C_DQT_ID_ERR_INT`: Triggered when the decoded quantization table ID of each component is different from the software configured quantization table ID.
- `JPEG_RST_UXP_ERR_INT`: Triggered when the `JPEG_RESTART_INTERVAL` configured by the software is 0 but the RST marker is parsed by the decoder.
- `JPEG_RST_CHECK_NONE_ERR_INT`: Triggered when the `JPEG_RESTART_INTERVAL` configured by the software is non-0 but the RST marker cannot be parsed by the decoder.
- `JPEG_RST_CHECK_POS_ERR_INT`: Triggered when the MCU number between two parsed RST markers is not equal to the `JPEG_RESTART_INTERVAL` configured by the software.
- `JPEG_OUT_EOF_INT`: Triggered when the EOF marker is read from the TX channel of 2D DMA.
- `JPEG_SR_COLOR_MODE_ERR_INT`: This interrupt is invalid.
- `JPEG_DCT_DONE_INT`: Triggered when the DCT or IDCT calculation of one data unit is completed.
- `JPEG_BS_LAST_BLOCK_EOF_INT`: Triggered when the encoding of the last data unit is completed.
- `JPEG_SCAN_CHECK_NONE_ERR_INT`: Triggered when an image frame has multiple scans to be decoded and the SOS marker is not parsed within (`JPEG_SOS_CHECK_BYTE_NUM` + 1) bytes in any scan header information.
- `JPEG_SCAN_CHECK_POS_ERR_INT`: Triggered when the position of the scan header information parsed by the decoder is wrong.
- `JPEG_UXP_DET_INT`: Triggered when the marker parsed by the decoder is not supported by the hardware.
- `JPEG_EN_FRAME_EOF_ERR_INT`: Triggered when the number of data units to be encoded read from 2D DMA is less than the number of data units calculated based on the image resolution configured by the software.
- `JPEG_EN_FRAME_EOF_LACK_INT`: Triggered when a frame of image to be encoded is completely read from 2D DMA but the EOF marker is not read.
- `JPEG_DE_FRAME_EOF_ERR_INT`: Triggered when the number of data units obtained after decoding a frame of image is different from the number of data units calculated based on the image resolution configured by the software.
- `JPEG_DE_FRAME_EOF_LACK_INT`: Triggered when the bitstream of a image is completely read from 2D DMA but the EOF marker or EOI marker is not read.
- `JPEG_SOS_UNMATCH_ERR_INT`: Triggered when the number of components in the scan header information parsed by the decoder is 0 or the header length in the scan header information parsed by the decoder does not match the actual header length.
- `JPEG_MARKER_ERR_FST_SCAN_INT`: Triggered when there is an error in the first scan header information parsed by the decoder. The errors include errors triggering `JPEG_CID_ERR_INT`, errors triggering `JPEG_C_DHT_DC_ID_ERR_INT`, errors triggering `JPEG_C_DHT_AC_ID_ERR_INT`, errors triggering `JPEG_C_DQT_ID_ERR_INT` and errors triggering `JPEG_SOS_UNMATCH_ERR_INT`.

- `JPEG_MARKER_ERR_OTHER_SCAN_INT`: Triggered when there is an error in the non-first scan header information parsed by the decoder. The error type is the same as the errors that trigger `JPEG_MARKER_ERR_FST_SCAN_INT`.
- `JPEG_UNDET_INT`: Triggered when the bitstream of an image is completely read from 2D DMA but the SOS marker is not read.
- `JPEG_DECODE_TIMEOUT_INT`: Triggered when the decoder is timeout. Section [35.7.2](#) describes how this interrupt will occur.

The above interrupt sources can only be triggered when the interrupt enable register `interrupt_source_name_ENA` is set to 1. Write 1 to `interrupt_source_name_CLR` will clear the interrupt.

**Note:**

For definitions of *interrupt*, *interrupt signal*, *interrupt source*, and their correlations, please refer to Chapter [12 Interrupt Matrix](#) > Section [12.2 Interrupt Terminology in ESP32-P4](#).

Each interrupt source can be configured by a common set of registers that are described in Section [Interrupt Configuration Registers](#). The specific registers can be found in Section [35.8 Register Summary](#).

## 35.7 Programming Procedures

### 35.7.1 JPEG Encoder

When the JPEG codec is working as an encoder, the configuration process is:

1. Activate the JPEG codec:
    - Set `HP_SYS_CLKRST_JPEG_SYS_CLK_EN` to 1 to enable the JPEG codec clock;
    - Set `HP_SYS_CLKRST_RST_EN_JPEG` to 0 to release the JPEG codec system level reset.
  2. Configure the JPEG codec:
    - Set `JPEG_SOFT_RST` to 1 to reset the JPEG codec, and then set `JPEG_SOFT_RST` to 0 to release the JPEG codec reset.
    - Set `JPEG_MODE` to 0 to configure the JPEG codec to be used as an encoder.
    - Select the original image format by configuring the following registers:
      - Format:
        - \* `JPEG_COLOR_SPACE`
        - \* `JPEG_EXTD_COLOR_SPACE_EN`
        - \* `JPEG_EXTD_COLOR_SPACE`
      - Pixel order: `JPEG_PIXEL_REV`
- See details in Table [35.5-1](#).
- Select whether to add the EOI marker (0xFFD9) at the end of the bitstream by configuring `JPEG_TAILER_EN`:

- 0: disable
- 1: enable
- Select the image format to be compressed by configuring [JPEG\\_SAMPLE\\_SEL](#):
  - 0: YUV444
  - 1: YUV422
  - 2: YUV420

Note that this register is valid only when the original image format is RGB, YUV444, or YUV422. See details in Table [35.5-1](#).

- Configure [JPEG\\_HA](#) as the original image width and [JPEG\\_VA](#) as the original image height.
  - Configure the quantization coefficient table by following the instructions in Section [35.5.1.3](#).
3. Configure 2D DMA (For detailed instructions or introduction to related concepts, please refer to Chapter [6 2D-DMA Controller \(2D-DMA\)](#).)

(a) Configure the linked list of 2D DMA TX channel.

- Set 2DEN to 1 to enable 2D function.
- Configure the Buffer address pointer as the starting address where the original image is stored.
- Configure pbyte to select the number of bytes of one pixel of the original image.
- Configure HA as the width of the original image and VA as the height of the original image.
- Configure hb as the number of pixels in the horizontal direction of an image block and vb as the number of pixels in the vertical direction of an image block.
- Configure mod as 1, which means continuous reading of image blocks of  $hb \times vb$  size.
- Configure (X, Y) as the coordinate of the first pixel in the starting image block. Among them, according to the original image format and the format of the image to be compressed, there are restrictions on hb and vb, as shown in the table [35.7-1](#):

**Table 35.7-1. JPEG Encoder hb and vb Configuration**

| Format of Original Image | Format of the Image to be Compressed | hb              | vb              |
|--------------------------|--------------------------------------|-----------------|-----------------|
| RGB/YUV444               | YUV444                               | Multiples of 8  | Multiples of 8  |
|                          | YUV422                               | Multiples of 16 | Multiples of 8  |
|                          | YUV420                               | Multiples of 16 | Multiples of 16 |
| YUV422                   | YUV422                               | Multiples of 16 | Multiples of 8  |
|                          | YUV420                               | Multiples of 16 | 16              |
| YUV420                   | YUV420                               | Multiples of 16 | 16              |
| GRAY                     | GRAY                                 | Multiples of 8  | Multiples of 8  |

(b) Configure the linked list of the 2D DMA RX channel:

- Set 2DEN to 0 to enable 1D function.



- Configure the Buffer address pointer as the starting address where the encoded bitstream will be stored.
  - Configure size as the number of bytes in the space to store the encoded bitstream.
  - configure mod as 0, which means one-time reading of image blocks of  $hb \times vb$  size.
- (c) Connect 2D DMA channels to JPEG codec:
- Connect the TX channel  $n$  to the JPEG codec by configuring `DMA2D_OUT_PERI_SEL_CHn` to 0;
  - Connect the RX channel  $n$  to the JPEG codec by configuring `DMA2D_IN_PERI_SEL_CHn` to 0.
- (d) Reset 2D DMA channels:
- 2D DMA TX channel  $n$ : Set `DMA2D_OUT_RST_CHn` to 1 to reset 2D DMA TX channel  $n$ , then set `DMA2D_OUT_RST_CHn` to 0 to release the reset of 2D DMA TX channel  $n$ .
  - 2D DMA RX channel  $n$ : Set `DMA2D_IN_RST_CHn` to 1 to reset 2D DMA RX channel  $n$ , then set `DMA2D_IN_RST_CHn` to 0 to release the reset of 2D DMA RX channel  $n$ .
- (e) Select the burst length of 2D DMA channels:
- Configure `DMA2D_OUT_MEM_BURST_LENGTH_CHn` to select the burst length of 2D DMA TX channel  $n$ .
  - Configure `DMA2D_IN_MEM_BURST_LENGTH_CHn` to select the burst length of 2D DMA RX channel  $n$ .
- (f) Select the 2D DMA output image block size by configuring `DMA2D_OUT_MACRO_BLOCK_SIZE_CHn` according to the format of original image and the format of the image to be compressed. See details in Table 35.7-2:
- 0: 8x8
  - 1: 16x8
  - 2: 16x16

Table 35.7-2. `DMA2D_OUT_MACRO_BLOCK_SIZE_CHn` Configuration

| Format of Original Image | Format of the Image to be Compressed | <code>DMA2D_OUT_MACRO_BLOCK_SIZE_CHn</code> |
|--------------------------|--------------------------------------|---------------------------------------------|
| RGB/YUV444               | YUV444                               | 0                                           |
|                          | YUV422                               | 1                                           |
|                          | YUV420                               | 2                                           |
| YUV422                   | YUV422                               | 1                                           |
|                          | YUV420                               | 2                                           |
| YUV420                   | YUV420                               | 2                                           |
| GRAY                     | GRAY                                 | 0                                           |

- (g) Enable the reorder function to improve bandwidth utilization by setting `DMA2D_OUT_REORDER_EN_CHO` to 1. Note that, this function is only available for 2D DMA TX channel0, not for any other TX channels.
- (h) Configure 2D DMA channel linked list address:
- 2D DMA TX channel: `DMA2D_OUTLINK_ADDR_CHn`

- 2D DMA RX channel: [DMA2D\\_INLINK\\_ADDR\\_CH \$n\$](#)
4. Enable interrupt sources:
    - [DMA2D\\_OUT\\_INT\\_ENA\\_CH \$n\$ \\_REG](#)
    - [DMA2D\\_IN\\_INT\\_ENA\\_CH \$n\$ \\_REG](#)
    - [JPEG\\_INT\\_ENA\\_REG](#)
  5. Start the transmission of 2D DMA channels:
    - 2D DMA TX channel: set [DMA2D\\_OUTLINK\\_START\\_CH \$n\$](#)  to 1 and then to 0
    - 2D DMA RX channel: set [DMA2D\\_INLINK\\_START\\_CH \$n\$](#)  to 1 and then to 0
  6. Start the JPEG codec encoding by setting the [JPEG\\_JPEG\\_START](#) field.
  7. Wait till [DMA2D\\_IN\\_SUC\\_EOF\\_CH \$n\$ \\_INT](#) becomes 1, indicating that the encoded bitstream has been completely written to the memory.

## 35.7.2 JPEG Decoder

When the JPEG codec is working as a decoder, the configuration process is:

1. Activate the JPEG codec:
  - Set [HP\\_SYS\\_CLKRST\\_JPEG\\_SYS\\_CLK\\_EN](#) to 1 to enable the JPEG codec clock
  - Set [HP\\_SYS\\_CLKRST\\_RST\\_EN\\_JPEG](#) to 0 to release the JPEG codec system level reset.
2. Configure the JPEG codec:
  - Set [JPEG\\_SOFT\\_RST](#) to 1 to reset the JPEG codec, and then set [JPEG\\_SOFT\\_RST](#) to 0 to release the JPEG codec reset.
  - Set [JPEG\\_MODE](#) to 1 to configure the JPEG codec to be used as a decoder.
  - Enable the RST marker check following the instructions in Section 35.5.2.2, if required.
  - Configure the quantization coefficient tables by following the instructions in Section 35.5.2.3.
  - Configure the Huffman tables by following the instructions in Section 35.5.2.4.
  - Configure the width and height of the decoded image:
    - Width: write width of the decoded image to [JPEG\\_HA](#).
    - Height: write the height of the decoded image to [JPEG\\_VA](#).
    - According to different decoded image formats, [JPEG\\_HA](#) and [JPEG\\_VA](#) must meet the requirements of Table 35.7-3.

**Table 35.7-3. JPEG Decoder [JPEG\\_HA](#) and [JPEG\\_VA](#) Configuration**

| Format of Decoded Image | <a href="#">JPEG_HA</a> | <a href="#">JPEG_VA</a> |
|-------------------------|-------------------------|-------------------------|
| YUV444                  | Multiples of 8          | 8                       |
| YUV422                  | Multiples of 16         | 8                       |
| YUV420                  | Multiples of 16         | 16                      |

|      |                |   |
|------|----------------|---|
| GRAY | Multiples of 8 | 8 |
|------|----------------|---|

3. Configure 2D DMA (For detailed instructions or introduction to related concepts, please refer to Chapter [6 2D-DMA Controller \(2D-DMA\)](#).)

(a) Configure the linked list of the 2D DMA TX channel.

- Set 2DEN to 0 to enable the 1D function.
- Configure the Buffer address pointer as the starting address where the bitstream to be decoded is stored.
- Configure the length and size as the number of bytes in the space to store the bitstream to be decoded.
- configure mod as 0, which means one-time reading of image blocks of hb x vb size.

(b) Configure the linked list of 2D DMA RX channel.

- Set 2DEN to 1 to enable 2D function.
- Configure the Buffer address pointer as the starting address where the decoded image will be stored.
- Configure pbyte to select the number of bytes of one pixel of the image which is decoded and has undergone 2D DMA color space conversion.
- Configure HA as the width of the output decoded image, configure VA as the height of the output decoded image, configure hb as the number of pixels in the horizontal direction of an image block, configure vb as the number of pixels in the vertical direction of an image block.
- Configure mod as 1, which means continuous reading of image blocks of hb × vb size.
- Configure (X, Y) as the coordinates of the first pixel in the starting image block. Among them, according to the decoded image format, there are restrictions on hb and vb, as shown in the table [35.7-4](#):

**Table 35.7-4. JPEG Decoder hb and vb Configuration**

| Decoded image format | hb              | vb              |
|----------------------|-----------------|-----------------|
| YUV444               | Multiples of 8  | Multiples of 8  |
| YUV422               | Multiples of 16 | Multiples of 8  |
| YUV420               | Multiples of 16 | Multiples of 16 |
| GRAY                 | Multiples of 8  | Multiples of 8  |

(c) Connect 2D DMA channels to JPEG codec:

- Connect the TX channel *n* to the JPEG codec by configuring [DMA2D\\_OUT\\_PERI\\_SEL\\_CHn](#) to 0;
- Connect the RX channel *n* to the JPEG codec by configuring [DMA2D\\_IN\\_PERI\\_SEL\\_CHn](#) to 0.

(d) Reset 2D DMA channels:

- 2D DMA TX channel *n*: set [DMA2D\\_OUT\\_RST\\_CHn](#) to 1 to reset 2D DMA TX channel *n*, then set [DMA2D\\_OUT\\_RST\\_CHn](#) to 0 to release the reset of 2D DMA TX channel *n*.

- 2D DMA RX channel $n$ : set `DMA2D_IN_RST_CH $n$`  to 1 to reset 2D DMA RX channel $n$ , then set `DMA2D_IN_RST_CH $n$`  to 0 to release the reset of 2D DMA RX channel $n$ .
- (e) Select the burst length of 2D DMA channels:
- Configure `DMA2D_OUT_MEM_BURST_LENGTH_CH $n$`  to select the burst length of 2D DMA TX channel $n$
  - Configure `DMA2D_IN_MEM_BURST_LENGTH_CH $n$`  to select the burst length of 2D DMA RX channel $n$
- (f) Select 2D DMA receive image block size by configuring `DMA2D_IN_MACRO_BLOCK_SIZE_CH $n$`  according to the format of the decoded image. See details in Table 35.7-5:
- 0: 8x8
  - 1: 16x8
  - 2: 16x16

Table 35.7-5. `DMA2D_IN_MACRO_BLOCK_SIZE_CH $n$`  Configuration

| Format of Decoded Image | <code>DMA2D_IN_MACRO_BLOCK_SIZE_CH<math>n</math></code> |
|-------------------------|---------------------------------------------------------|
| YUV444                  | 0                                                       |
| YUV422                  | 1                                                       |
| YUV420                  | 2                                                       |
| GRAY                    | 0                                                       |

- (g) Enable the reorder function to improve bandwidth utilization by setting set `DMA2D_IN_REORDER_EN_CHO` to 1. Note that, this function is only available for 2D DMA RX channel0, not for any other RX channels.
- (h) Select whether to scramble the pixel order of the decoded image before 2D DMA color space conversion by configuring `DMA2D_IN_SCRAMBLE_SEL_PRE_CHO`:
- 0: BYTE2-1-0
  - 1: BYTE2-0-1
  - 2: BYTE1-0-2
  - 3: BYTE1-2-0
  - 4: BYTE0-2-1
  - 5: BYTE0-1-2

Configuring `DMA2D_IN_SCRAMBLE_SEL_PRE_CHO` to 5 means swapping the pixel order, for example, swapping YUV444 output by the JPEG decoder to VUY444. Note that, this function is only available for 2D DMA RX channel0, not for any other RX channels.

- (i) Select the color conversion methods by configuring `DMA2D_IN_COLOR_INPUT_SEL_CHO`, `DMA2D_IN_COLOR_3B_PROC_EN_CHO`, and `DMA2D_IN_COLOR_OUTPUT_SEL_CHO`. See details in Table 35.7-6:

Table 35.7-6. 2D DMA Color Space Conversion Configuration

| Color Space Conversion Method         | INPUT_SEL_CHO | 3B_PROC_EN_CHO | OUTPUT_SEL_CHO |
|---------------------------------------|---------------|----------------|----------------|
| YUV420 → RGB                          | 0             | 1              | 1              |
| YUV422 → RGB                          | 0             | 1              | 1              |
| YUV444 → RGB                          | 2             | 1              | 1              |
| YUV444 (only reverse the pixel order) | 2             | 0              | 1              |
| No color space conversion             | 7             | N/A            | N/A            |

Note that, this function is only available for 2D DMA RX channel0, not for any other RX channels.

- (j) Configure parameters required for the color space conversion formula, including DMA2D\_IN\_COLOR\_PARAM\_H0\_CHO, DMA2D\_IN\_COLOR\_PARAM\_H1\_CHO, DMA2D\_IN\_COLOR\_PARAM\_M0\_CHO, DMA2D\_IN\_COLOR\_PARAM\_M1\_CHO, DMA2D\_IN\_COLOR\_PARAM\_LO\_CHO, and DMA2D\_IN\_COLOR\_PARAM\_L1\_CHO. Note that, this function is only available for 2D DMA RX channel0, not for any other RX channels.
- (k) Select whether to reverse the pixel order of the image after 2D DMA color space conversion by configuring DMA2D\_IN\_SCRAMBLE\_SEL\_POST\_CHO:
- 0: BYTE2-1-0
  - 1: BYTE2-0-1
  - 2: BYTE1-0-2
  - 3: BYTE1-2-0
  - 4: BYTE0-2-1
  - 5: BYTE0-1-2

Configuration of 5 means reversing the pixel order, for example, reversing the RGB after 2D DMA color space conversion to BGR. Note that, this function is only available for 2D DMA RX channel0, not for any other RX channels.

- (l) Configure [DMA2D\\_OUTLINK\\_ADDR\\_CHn](#) as the 2D DMA TX channel linked list address, configure [DMA2D\\_INLINK\\_ADDR\\_CHn](#) as the 2D DMA RX channel linked list address.

4. Enable interrupt sources:

- [DMA2D\\_OUT\\_INT\\_ENA\\_CHn\\_REG](#)
- [DMA2D\\_IN\\_INT\\_ENA\\_CHn\\_REG](#)
- [JPEG\\_INT\\_ENA\\_REG](#)

5. Start the transmission of 2D DMA channels:

- 2D DMA TX channeln: set [DMA2D\\_OUTLINK\\_START\\_CHn](#) to 1 and then to 0
- 2D DMA RX channeln: set [DMA2D\\_INLINK\\_START\\_CHn](#) to 1 and then to 0

6. Start JPEG codec decoding by setting the [JPEG\\_JPEG\\_START](#) field.

7. Wait till DMA2D\_IN\_SUC\_EOF\_CHn\_INT becomes 1, indicating that the decoded image has been completely written to the memory.

### 35.7.3 Reset

Users can **reset** JPEG codec entirely or partially, depending on the actual scenarios, at any time during the encoding or decoding process:

- The entire JPEG codec, including all the RAMs and FIFOs as well as all the state machines: set [JPEG\\_SOFT\\_RST](#) to 1. In this case, the register configuration value will not be reset.
- Only all the RAMs and FIFOs inside the JPEG codec, for example, changing the quantization coefficient tables or Huffman tables: set [JPEG\\_FIFO\\_RST](#) to 1.
- Only all the state machines inside the JPEG codec, for example, only stop the current encoding or decoding process: set [JPEG\\_FSM\\_RST](#) to 1.

## 35.8 Register Summary

The addresses in this section are relative to JPEG Codec base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

| Name                                       | Description                                            | Address | Access   |
|--------------------------------------------|--------------------------------------------------------|---------|----------|
| <b>Control and Configuration registers</b> |                                                        |         |          |
| JPEG_CONFIG_REG                            | System level control and configuration register        | 0x0000  | varies   |
| JPEG_DQT_INFO_REG                          | Quantization coefficient table information register    | 0x0004  | R/W      |
| JPEG_PIC_SIZE_REG                          | Image size register                                    | 0x0008  | R/W      |
| JPEG_EXTD_CONFIG_REG                       | Extended color space conversion configuration register | 0x000C  | R/W      |
| JPEG_TOQNR_REG                             | Quantization coefficient table0 register               | 0x0010  | HRO      |
| JPEG_T1QNR_REG                             | Quantization coefficient table1 register               | 0x0014  | HRO      |
| JPEG_T2QNR_REG                             | Quantization coefficient table2 register               | 0x0018  | HRO      |
| JPEG_T3QNR_REG                             | Quantization coefficient table3 register               | 0x001C  | HRO      |
| JPEG_DECODE_CONF_REG                       | Decoder configuration register                         | 0x0020  | varies   |
| JPEG_CO_REG                                | Color component0 register                              | 0x0024  | R/W      |
| JPEG_C1_REG                                | Color component1 register                              | 0x0028  | R/W      |
| JPEG_C2_REG                                | Color component2 register                              | 0x002C  | R/W      |
| JPEG_C3_REG                                | Color component3 register                              | 0x0030  | R/W      |
| JPEG_DHT_INFO_REG                          | Huffman table information register                     | 0x0034  | R/W      |
| JPEG_DHT_TOTLEN_DCO_REG                    | DC0 Huffman table codeword length register             | 0x0058  | HRO      |
| JPEG_DHT_VAI_DCO_REG                       | DC0 Huffman table symbol register                      | 0x005C  | HRO      |
| JPEG_DHT_TOTLEN_ACO_REG                    | ACO Huffman table codeword length register             | 0x0060  | HRO      |
| JPEG_DHT_VAI_ACO_REG                       | ACO Huffman table symbol register                      | 0x0064  | HRO      |
| JPEG_DHT_TOTLEN_DC1_REG                    | DC1 Huffman table codeword length register             | 0x0068  | HRO      |
| JPEG_DHT_VAI_DC1_REG                       | DC1 Huffman table symbol register                      | 0x006C  | HRO      |
| JPEG_DHT_TOTLEN_AC1_REG                    | AC1 Huffman table codeword length register             | 0x0070  | HRO      |
| JPEG_DHT_VAI_AC1_REG                       | AC1 Huffman table symbol register                      | 0x0074  | HRO      |
| JPEG_DHT_CODEMIN_DCO_REG                   | DC0 Huffman table minimum codeword register            | 0x0078  | HRO      |
| JPEG_DHT_CODEMIN_ACO_REG                   | ACO Huffman table minimum codeword register            | 0x007C  | HRO      |
| JPEG_DHT_CODEMIN_DC1_REG                   | DC1 Huffman table minimum codeword register            | 0x0080  | HRO      |
| JPEG_DHT_CODEMIN_AC1_REG                   | AC1 Huffman table minimum codeword register            | 0x0084  | HRO      |
| JPEG_SYS_REG                               | System configuration register                          | 0x00F8  | R/W      |
| <b>Interrupt registers</b>                 |                                                        |         |          |
| JPEG_INT_RAW_REG                           | Interrupt raw status register                          | 0x0038  | R/WTC/SS |
| JPEG_INT_ENA_REG                           | Interrupt enable register                              | 0x003C  | R/W      |
| JPEG_INT_ST_REG                            | Interrupt masked status register                       | 0x0040  | RO       |
| JPEG_INT_CLR_REG                           | Interrupt clear register                               | 0x0044  | WT       |
| <b>Version Register</b>                    |                                                        |         |          |
| JPEG_VERSION_REG                           | Version control register                               | 0x00FC  | R/W      |

## 35.9 Registers

The addresses in this section are relative to the JPEG Codec base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section [Programming Reserved Register Field](#).

### Register 35.1. JPEG\_CONFIG\_REG (0x0000)

| JPEG_MODE<br>(reserved) |    |    |    |    |    |    |    |    |    | JPEG_DECODE_TIMEOUT_TASK_SEL |    |    |    |    |    |    |    |   |   | JPEG_DECODE_TIMEOUT_THRES |   |   |   |   |   |   |   |   |       | JPEG_MEM_CLK_FORCE_ON |  |  |  |  |  |  |  |  |  | JPEG_DHT_FIFO_EN |  |  |  |  |  |  |  |  |  | JPEG_COLOR_SPACE |  |  |  |  |  |  |  |  |  | JPEG_CNQR_TBL_SEL |  |  |  |  |  |  |  |  |  | JPEG_LQNR_TBL_SEL |  |  |  |  |  |  |  |  |  | JPEG_ONR_FIFO_EN |  |  |  |  |  |  |  |  |  | JPEG_DEBUG_DIRECT_OUT_EN |  |  |  |  |  |  |  |  |  | JPEG_DMA_LINKLIST_MODE |  |  |  |  |  |  |  |  |  | JPEG_SAMPLE_SEL |  |  |  |  |  |  |  |  |  | JPEG_FF_CHECK_EN |  |  |  |  |  |  |  |  |  | JPEG_ONR_PRECISION |  |  |  |  |  |  |  |  |  | JPEG_START |  |  |  |  |  |  |  |  |  | JPEG_FSM_RST |  |  |  |  |  |  |  |  |  |
|-------------------------|----|----|----|----|----|----|----|----|----|------------------------------|----|----|----|----|----|----|----|---|---|---------------------------|---|---|---|---|---|---|---|---|-------|-----------------------|--|--|--|--|--|--|--|--|--|------------------|--|--|--|--|--|--|--|--|--|------------------|--|--|--|--|--|--|--|--|--|-------------------|--|--|--|--|--|--|--|--|--|-------------------|--|--|--|--|--|--|--|--|--|------------------|--|--|--|--|--|--|--|--|--|--------------------------|--|--|--|--|--|--|--|--|--|------------------------|--|--|--|--|--|--|--|--|--|-----------------|--|--|--|--|--|--|--|--|--|------------------|--|--|--|--|--|--|--|--|--|--------------------|--|--|--|--|--|--|--|--|--|------------|--|--|--|--|--|--|--|--|--|--------------|--|--|--|--|--|--|--|--|--|
| 31                      | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 17                           | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7                         | 6 | 5 | 4 | 3 | 2 | 1 | 0 |   |       |                       |  |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  |  |  |                          |  |  |  |  |  |  |  |  |  |                        |  |  |  |  |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  |  |  |                    |  |  |  |  |  |  |  |  |  |            |  |  |  |  |  |  |  |  |  |              |  |  |  |  |  |  |  |  |  |
| 0                       | 00 | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 32 | 0                            | 1  | 0  | 1  | 0  | 1  | 0  | 1  | 0 | 1 | 0                         | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | Reset |                       |  |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  |  |  |                          |  |  |  |  |  |  |  |  |  |                        |  |  |  |  |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  |  |  |                    |  |  |  |  |  |  |  |  |  |            |  |  |  |  |  |  |  |  |  |              |  |  |  |  |  |  |  |  |  |

**JPEG\_FSM\_RST** Configures whether or not to reset the state machine of JPEG Codec.

0: Invalid. No effect

1: Reset the state machine

(WT)

**JPEG\_JPEG\_START** Configures whether or not to start compressing a new image.

0: Invalid. No effect

1: Start compressing a new image

(WT)

**JPEG\_QNR\_PRECISION** Configures the quantization coefficient table precision for the encoder.

0: 8-bit precision

1: 16-bit precision

(R/W)

**JPEG\_FF\_CHECK\_EN** Configures whether or not to add “0x00” after “0xFF”.

0: Not add

1: Add

(R/W)

**JPEG\_SAMPLE\_SEL** Configures the format of the image to be compressed.

O: YUV444

1: YUV422

2: YUV420

3: Invalid. No effect

(R/W)

Continued on the next page...



**Register 35.1. JPEG\_CONFIG\_REG (0x0000)**

Continued from the previous page...

**JPEG\_DMA\_LINKLIST\_MODE** Represents 2D DMA linked list mode.

0: Invalid. No effect

1: 2D DMA uses linked lists to configure

(RO)

**JPEG\_DEBUG\_DIRECT\_OUT\_EN** Configures whether or not to enable debug mode for encoder.

0: Disable debug mode, the encoder will work in normal mode

1: Enable debug mode, the input image will be directly output by the encoder

(R/W)

**JPEG\_QNR\_FIFO\_EN** Configures whether or not to enable FIFO mode when configuring quantization coefficient tables.

0: Disable. Use non-FIFO mode

1: Enable

(R/W)

**JPEG\_LQNR\_TBL\_SEL** Configures the luminance quantization table ID for the encoder. (R/W)

**JPEG\_CQNR\_TBL\_SEL** Configures the chrominance quantization table ID for the encoder. (R/W)

**JPEG\_COLOR\_SPACE** Configures the original image's color space.

0: RGB888

1: YUV422

2: RGB565

3: GRAY

(R/W)

**JPEG\_DHT\_FIFO\_EN** Configures whether or not to enable FIFO mode when configuring Huffman tables.

0: Disable. Use non-FIFO mode

1: Enable

Note that to read Huffman tables only non-FIFO mode is supported.

(R/W)

**JPEG\_MEM\_CLK\_FORCE\_ON** Configures whether or not to force on memory's clock gate.

0: Not force on

1: Force on

(R/W)

**JPEG\_DECODE\_TIMEOUT\_THRES** Configures decode timeout period to trigger [JPEG\\_DECODE\\_TIMEOUT\\_INT\\_RAW](#). The clock cycles of timeout period =  $2^{JPEG\_DECODE\_TIMEOUT\_THRES} - 1$ . (R/W)

Continued on the next page...

**Register 35.1. JPEG\_CONFIG\_REG (0x0000)**

Continued from the previous page...

**JPEG\_DECODE\_TIMEOUT\_TASK\_SEL** Configures the reset mode upon the decoder timeout.

0: Software uses reset to abort the decoding process

1: Hardware automatically aborts decoding process

(R/W)

**JPEG\_SOFT\_RST** Configures whether or not to apply soft reset to the JPEG Codec.

0: Release the soft reset

1: Apply soft reset to the JPEG Codec (the register configuration value will not be reset)

(R/W)

**JPEG\_FIFO\_RST** Configures whether or not to apply FIFO reset to the JPEG Codec.

0: Release the FIFO reset

1: Apply FIFO reset to the JPEG Codec

(R/W)

**JPEG\_PIXEL\_REV** Configures whether or not to reverse the original image's pixel order.

0: Not reverse

1: Reverse

(R/W)

**JPEG\_TAILER\_EN** Configures whether or not to add EOI marker "0xFFD9" at the end of bitstream.

0: Not add

1: Add

(R/W)

**JPEG\_PAUSE\_EN** Configures whether or not to pause the JPEG Codec.

0: Not pause

1: Pause

(R/W)

**JPEG\_MODE** Configures if the JPEG Codec is working as an encoder or a decoder.

0: Encoder

1: Decoder

(R/W)

Register 35.2. JPEG\_DQT\_INFO\_REG (0x0004)

|                  |    |    |    |    |   |   |   |
|------------------|----|----|----|----|---|---|---|
| JPEG_T3_DQT_INFO |    |    |    |    |   |   |   |
| JPEG_T2_DQT_INFO |    |    |    |    |   |   |   |
| JPEG_T1_DQT_INFO |    |    |    |    |   |   |   |
| JPEG_TO_DQT_INFO |    |    |    |    |   |   |   |
| 31               | 24 | 23 | 16 | 15 | 8 | 7 | 0 |
| 3                |    | 2  |    | 1  |   | 0 |   |
| Reset            |    |    |    |    |   |   |   |

**JPEG\_TO\_DQT\_INFO** Configures quantization coefficient table0's precision and ID for decoder.

Bit[7:4]: Configures precision

Bit[3:0]: Configures ID

(R/W)

**JPEG\_T1\_DQT\_INFO** Configures quantization coefficient table1's precision and ID for decoder. See details in [JPEG\\_TO\\_DQT\\_INFO](#). (R/W)

**JPEG\_T2\_DQT\_INFO** Configures quantization coefficient table2's precision and ID for decoder. See details in [JPEG\\_TO\\_DQT\\_INFO](#). (R/W)

**JPEG\_T3\_DQT\_INFO** Configures quantization coefficient table3's precision and ID for decoder. See details in [JPEG\\_TO\\_DQT\\_INFO](#). (R/W)

Register 35.3. JPEG\_PIC\_SIZE\_REG (0x0008)

|         |  |  |  |  |  |  |  |     |  |  |  |  |  |  |       |
|---------|--|--|--|--|--|--|--|-----|--|--|--|--|--|--|-------|
| JPEG_HA |  |  |  |  |  |  |  |     |  |  |  |  |  |  |       |
| JPEG_VA |  |  |  |  |  |  |  |     |  |  |  |  |  |  |       |
| 31      |  |  |  |  |  |  |  |     |  |  |  |  |  |  | 0     |
| 640     |  |  |  |  |  |  |  | 480 |  |  |  |  |  |  |       |
|         |  |  |  |  |  |  |  |     |  |  |  |  |  |  | Reset |

**JPEG\_VA** Configures the image's height.

When JPEG Codec works as an encoder, the maximum configurable bits is 14.

When JPEG Codec works as a decoder, the maximum configurable bits is 16.

(R/W)

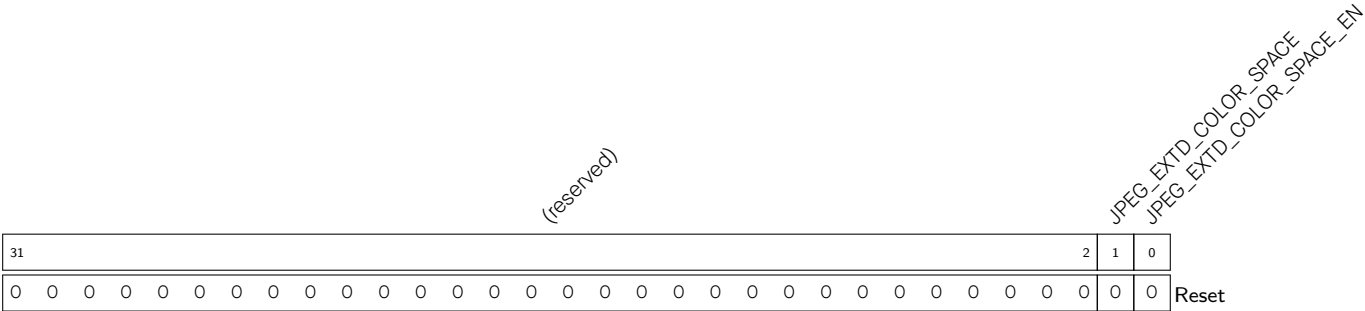
**JPEG\_HA** Configures the image's width.

When JPEG Codec works as an encoder, the maximum configurable bits is 14.

When JPEG Codec works as a decoder, the maximum configurable bits is 16.

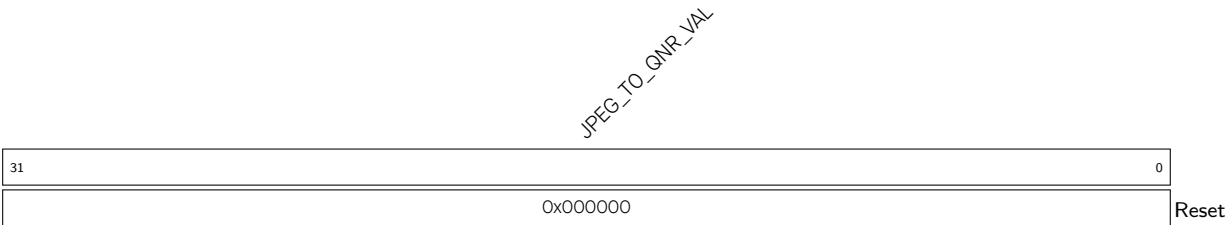
(R/W)

Register 35.4. JPEG\_EXTD\_CONFIG\_REG (0x000C)



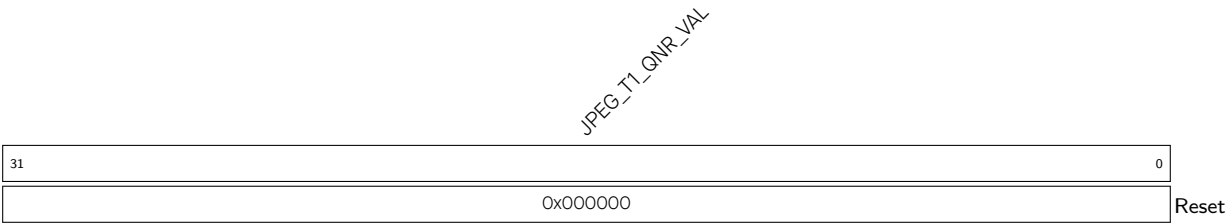
- JPEG\_EXTD\_COLOR\_SPACE\_EN   Configure whether to enable extended color space conversion options.  
0: Disable  
1: Enable  
(R/W)
- JPEG\_EXTD\_COLOR\_SPACE   Configure the extended color space format to be converted. Valid when JPEG\_EXTD\_COLOR\_SPACE\_EN configured to 1.  
0: YUV444  
1: YUV420  
(R/W)

Register 35.5. JPEG\_TOQNR\_REG (0x0010)

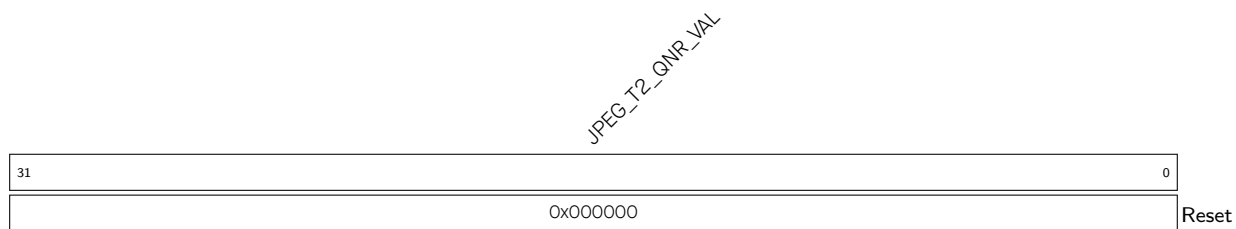


- JPEG\_TO\_QNR\_VAL   Configures the quantization coefficients of the quantization table 0 in FIFO mode. (HRO)

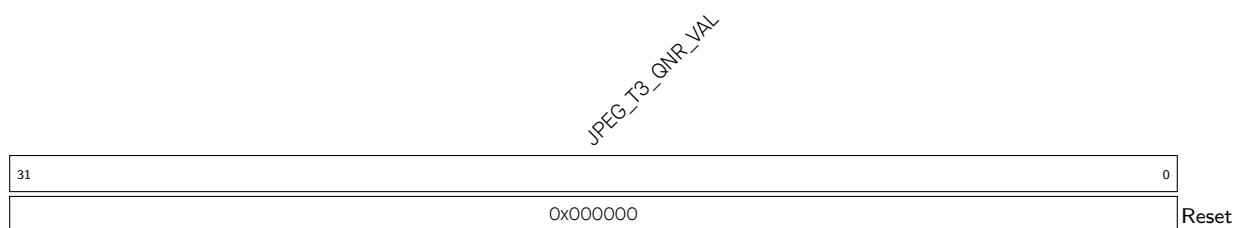
Register 35.6. JPEG\_T1QNR\_REG (0x0014)



- JPEG\_T1\_QNR\_VAL   Configures the quantization coefficients of the quantization table 1 in FIFO mode. (HRO)

**Register 35.7. JPEG\_T2QNR\_REG (0x0018)**

**JPEG\_T2\_QNR\_VAL** Configures the quantization coefficients of the quantization table 2 in FIFO mode. (HRO)

**Register 35.8. JPEG\_T3QNR\_REG (0x001C)**

**JPEG\_T3\_QNR\_VAL** Configures the quantization coefficients of the quantization table 3 in FIFO mode. (HRO)

Register 35.9. JPEG\_DECODE\_CONF\_REG (0x0020)

|            |    |    |    |    |    |    |    |    |    |  |  |  |  |  |  |                         |  |  |  |  |  |      |  |  |    |   |  |  |  |  |  |                           |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                         |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                         |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                |  |  |  |  |       |  |  |  |  |  |  |  |  |  |  |                    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|------------|----|----|----|----|----|----|----|----|----|--|--|--|--|--|--|-------------------------|--|--|--|--|--|------|--|--|----|---|--|--|--|--|--|---------------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|-------------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|-------------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|----------------|--|--|--|--|-------|--|--|--|--|--|--|--|--|--|--|--------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|-----------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
| (reserved) |    |    |    |    |    |    |    |    |    |  |  |  |  |  |  | JPEG_DEZIGZAG_READY_CTL |  |  |  |  |  |      |  |  |    |   |  |  |  |  |  | JPEG_MULTI_SCAN_ERR_CHECK |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | JPEG_RST_CHECK_BYTE_NUM |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | JPEG_SOS_CHECK_BYTE_NUM |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | JPEG_SW_DHT_EN |  |  |  |  |       |  |  |  |  |  |  |  |  |  |  | JPEG_COMPONENT_NUM |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | JPEG_RESTART_INTERVAL |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 31         | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 16 |  |  |  |  |  |  |                         |  |  |  |  |  |      |  |  | 15 | 0 |  |  |  |  |  |                           |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                         |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                         |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                |  |  |  |  |       |  |  |  |  |  |  |  |  |  |  |                    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 0          | 1  | 0  | 3  | 3  | 1  | 3  |    |    |    |  |  |  |  |  |  |                         |  |  |  |  |  | 0x00 |  |  |    |   |  |  |  |  |  |                           |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                         |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                         |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                |  |  |  |  | Reset |  |  |  |  |  |  |  |  |  |  |                    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

Reset

**JPEG\_RESTART\_INTERVAL** Configures the RST interval in DRI segment when decoding. (R/W)

**JPEG\_COMPONENT\_NUM** Configures the number of chrominance components in the image when decoding. (R/W)

**JPEG\_SW\_DHT\_EN** Represents the Huffman table configured by the software for the decoder. The value is always 1. (RO)

**JPEG\_SOS\_CHECK\_BYTE\_NUM** Configures the byte number to check the next SOS marker in the multi-scan image after one scan is decoded down. The actual check number is [JPEG\\_SOS\\_CHECK\\_BYTE\\_NUM](#) + 1 (R/W)

**JPEG\_RST\_CHECK\_BYTE\_NUM** Configures the byte number to check the next RST marker after one RST interval is decoded down. The actual check number is [JPEG\\_RST\\_CHECK\\_BYTE\\_NUM](#) + 1. (R/W)

**JPEG\_MULTI\_SCAN\_ERR\_CHECK** Reserved for decoder and should not be configured. (R/W)

**JPEG\_DEZIGZAG\_READY\_CTL** Reserved for decoder and should not be configured. (R/W)

**Register 35.10. JPEG\_CO\_REG (0x0024)**

|            |    |    |    |    |    |    |   |            |   |       |   |   |   |  |  |                  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  |                     |  |  |  |  |  |  |  |
|------------|----|----|----|----|----|----|---|------------|---|-------|---|---|---|--|--|------------------|--|--|--|--|--|--|--|------------------|--|--|--|--|--|--|--|---------------------|--|--|--|--|--|--|--|
| (reserved) |    |    |    |    |    |    |   | JPEG_CO_ID |   |       |   |   |   |  |  | JPEG_CO_X_FACTOR |  |  |  |  |  |  |  | JPEG_CO_Y_FACTOR |  |  |  |  |  |  |  | JPEG_CO_DQT_TBL_SEL |  |  |  |  |  |  |  |
| 31         | 24 | 23 | 16 | 15 | 12 | 11 | 8 | 7          | 0 |       |   |   |   |  |  |                  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  |                     |  |  |  |  |  |  |  |
| 0          | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0          | 0 | 0     | 1 | 1 | 0 |  |  |                  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  |                     |  |  |  |  |  |  |  |
|            |    |    |    |    |    |    |   |            |   | Reset |   |   |   |  |  |                  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  |                     |  |  |  |  |  |  |  |

Reset

**JPEG\_CO\_DQT\_TBL\_SEL** Configures the selected quantization coefficient table ID for component0 in decoder mode. (R/W)

**JPEG\_CO\_Y\_FACTOR** Configures the vertical sampling factor of component0 in decoder mode. (R/W)

**JPEG\_CO\_X\_FACTOR** Configures the horizontal sampling factor of component0 in decoder mode. (R/W)

**JPEG\_CO\_ID** Configures the ID of component0 in decoder mode. (R/W)

**Register 35.11. JPEG\_C1\_REG (0x0028)**

|            |  |  |  |  |  |  |  |            |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  |                     |  |  |  |  |  |  |  |    |  |  |  |  |  |  |  |    |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  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| 31         |  |  |  |  |  |  |  | 24         |  |  |  |  |  |  |  | 23               |  |  |  |  |  |  |  | 16               |  |  |  |  |  |  |  | 15                  |  |  |  |  |  |  |  | 12 |  |  |  |  |  |  |  | 11 |  |  |  |  |  |  |  | 8 |  |  |  |  |  |  |  | 7 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  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| 0          |  |  |  |  |  |  |  | 0          |  |  |  |  |  |  |  | 0                |  |  |  |  |  |  |  | 0                |  |  |  |  |  |  |  | 0                   |  |  |  |  |  |  |  | 0  |  |  |  |  |  |  |  | 0  |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  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|  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  |

Reset

**JPEG\_C1\_DQT\_TBL\_SEL** Configures the selected quantization coefficient table ID for component1 in decoder mode. (R/W)

**JPEG\_C1\_Y\_FACTOR** Configures the vertical sampling factor of component1 in decoder mode. (R/W)

**JPEG\_C1\_X\_FACTOR** Configures the horizontal sampling factor of component1 in decoder mode. (R/W)

**JPEG\_C1\_ID** Configures the ID of component1 in decoder mode. (R/W)

**Register 35.12. JPEG\_C2\_REG (0x002C)**

|            |    |    |    |    |    |    |   |            |   |   |   |   |   |   |   |                  |   |   |   |   |   |   |   |                  |   |   |   |   |   |   |   |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|------------|----|----|----|----|----|----|---|------------|---|---|---|---|---|---|---|------------------|---|---|---|---|---|---|---|------------------|---|---|---|---|---|---|---|---------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| (reserved) |    |    |    |    |    |    |   | JPEG_C2_ID |   |   |   |   |   |   |   | JPEG_C2_X_FACTOR |   |   |   |   |   |   |   | JPEG_C2_Y_FACTOR |   |   |   |   |   |   |   | JPEG_C2_DQT_TBL_SEL |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 31         | 24 | 23 | 16 | 15 | 12 | 11 | 8 | 7          | 0 |   |   |   |   |   |   |                  |   |   |   |   |   |   |   |                  |   |   |   |   |   |   |   |                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0          | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0          | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0                | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0                | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0                   | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

Reset

**JPEG\_C2\_DQT\_TBL\_SEL** Configures the selected quantization coefficient table ID for component2 in decoder mode. (R/W)

**JPEG\_C2\_Y\_FACTOR** Configures the vertical sampling factor of component2 in decoder mode. (R/W)

**JPEG\_C2\_X\_FACTOR** Configures the horizontal sampling factor of component2 in decoder mode. (R/W)

**JPEG\_C2\_ID** Configures the ID of component2 in decoder mode. (R/W)

**Register 35.13. JPEG\_C3\_REG (0x0030)**

|            |  |  |  |  |  |  |  |            |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  |                  |  |  |  |  |  |  |  |                     |  |  |  |  |  |  |  |    |  |  |  |  |  |  |  |    |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  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| 31         |  |  |  |  |  |  |  | 24         |  |  |  |  |  |  |  | 23               |  |  |  |  |  |  |  | 16               |  |  |  |  |  |  |  | 15                  |  |  |  |  |  |  |  | 12 |  |  |  |  |  |  |  | 11 |  |  |  |  |  |  |  | 8 |  |  |  |  |  |  |  | 7 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |   |  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| 0          |  |  |  |  |  |  |  | 0          |  |  |  |  |  |  |  | 0                |  |  |  |  |  |  |  | 0                |  |  |  |  |  |  |  | 0                   |  |  |  |  |  |  |  | 0  |  |  |  |  |  |  |  | 0  |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  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|  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  |

Reset

**JPEG\_C3\_DQT\_TBL\_SEL** Configures the selected quantization coefficient table ID for component3 in decoder mode. (R/W)

**JPEG\_C3\_Y\_FACTOR** Configures the vertical sampling factor of component3 in decoder mode. (R/W)

**JPEG\_C3\_X\_FACTOR** Configures the horizontal sampling factor of component3 in decoder mode. (R/W)

**JPEG\_C3\_ID** Configures the ID of component3 in decoder mode. (R/W)



Register 35.14. JPEG\_DHT\_INFO\_REG (0x0034)

|                  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                 |   |   |   |                 |  |                 |  |                 |  |
|------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|-----------------|---|---|---|-----------------|--|-----------------|--|-----------------|--|
| (reserved)       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | JPEG_AC1_DHT_ID |   |   |   | JPEG_ACO_DHT_ID |  | JPEG_DC1_DHT_ID |  | JPEG_DCO_DHT_ID |  |
| 311615121187430  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                 |   |   |   |                 |  |                 |  |                 |  |
| 0000000000000000 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 1               | 0 | 1 | 0 | Reset           |  |                 |  |                 |  |

Reset

**JPEG\_DCO\_DHT\_ID** Configures the ID of DCO Huffman table in decoder mode. (R/W)

**JPEG\_DC1\_DHT\_ID** Configures the ID of DC1 Huffman table in decoder mode. (R/W)

**JPEG\_ACO\_DHT\_ID** Configures the ID of ACO Huffman table in decoder mode. (R/W)

**JPEG\_AC1\_DHT\_ID** Configures the ID of AC1 Huffman table in decoder mode. (R/W)

Register 35.15. JPEG\_INT\_RAW\_REG (0x0038)

|                                    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                   |
|------------------------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|-------------------|
| (reserved)                         |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | JPEG_DONE_INT_RAW |
| JPEG_DECODE_TIMEOUT_INT_RAW        |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                   |
| JPEG_UNDET_INT_RAW                 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                   |
| JPEG_MARKER_ERR_OTHER_SCAN_INT_RAW |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                   |
| JPEG_MARKER_ERR_EST_SCAN_INT_RAW   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                   |
| JPEG_SOS_UNMATCH_ERR_INT_RAW       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                   |
| JPEG_DE_FRAME_EOF_LACK_INT_RAW     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                   |
| JPEG_EN_FRAME_EOF_LACK_INT_RAW     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                   |
| JPEG_EN_FRAME_EOF_ERR_INT_RAW      |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                   |
| JPEG_UXP_DET_INT_RAW               |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                   |
| JPEG_SCAN_CHECK_POS_ERR_INT_RAW    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                   |
| JPEG_SCAN_CHECK_NONE_ERR_INT_RAW   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                   |
| JPEG_BS_LAST_BLOCK_INT_RAW         |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                   |
| JPEG_DCT_DONE_INT_RAW              |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                   |
| JPEG_SR_COLOR_MODE_ERR_INT_RAW     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                   |
| JPEG_OUT_EOF_INT_RAW               |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                   |
| JPEG_RST_CHECK_POS_ERR_INT_RAW     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                   |
| JPEG_RST_CHECK_NONE_ERR_INT_RAW    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                   |
| JPEG_C_DQT_ID_ERR_INT_RAW          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                   |
| JPEG_C_DHT_DC_ID_ERR_INT_RAW       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                   |
| JPEG_C_DHT_AC_ID_ERR_INT_RAW       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                   |
| JPEG_OID_ERR_INT_RAW               |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                   |
| JPEG_RST_PARALLEL_ERR_INT_RAW      |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                   |
| JPEG_DONE_INT_RAW                  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                   |

|    |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |       |
|----|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|-------|
| 31 |   |   |   |   |   |   | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |       |
| 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | Reset |

Reset

**JPEG\_DONE\_INT\_RAW** The raw status of JPEG\_DONE\_INT. (R/WTC/SS)

**JPEG\_RLE\_PARALLEL\_ERR\_INT\_RAW** The raw status of JPEG\_RLE\_PARALLEL\_ERR\_INT. (R/WTC/SS)

**JPEG\_CID\_ERR\_INT\_RAW** The raw status of JPEG\_CID\_ERR\_INT. (R/WTC/SS)

**JPEG\_C\_DHT\_DC\_ID\_ERR\_INT\_RAW** The raw status of JPEG\_C\_DHT\_DC\_ID\_ERR\_INT. (R/WTC/SS)

**JPEG\_C\_DHT\_AC\_ID\_ERR\_INT\_RAW** The raw status of JPEG\_C\_DHT\_AC\_ID\_ERR\_INT. (R/WTC/SS)

**JPEG\_C\_DQT\_ID\_ERR\_INT\_RAW** The raw status of JPEG\_C\_DQT\_ID\_ERR\_INT. (R/WTC/SS)

**JPEG\_RST\_UXP\_ERR\_INT\_RAW** The raw status of JPEG\_RST\_UXP\_ERR\_INT. (R/WTC/SS)

**JPEG\_RST\_CHECK\_NONE\_ERR\_INT\_RAW** The raw status of JPEG\_RST\_CHECK\_NONE\_ERR\_INT. (R/WTC/SS)

Continued on the next page...

**Register 35.15. JPEG\_INT\_RAW\_REG (0x0038)**

Continued from the previous page...

**JPEG\_RST\_CHECK\_POS\_ERR\_INT\_RAW** The raw status of JPEG\_RST\_CHECK\_POS\_ERR\_INT.  
(R/WTC/SS)

**JPEG\_OUT\_EOF\_INT\_RAW** The raw status of JPEG\_OUT\_EOF\_INT. (R/WTC/SS)

**JPEG\_SR\_COLOR\_MODE\_ERR\_INT\_RAW** The raw status of JPEG\_SR\_COLOR\_MODE\_ERR\_INT.  
This interrupt is invalid. (R/WTC/SS)

**JPEG\_DCT\_DONE\_INT\_RAW** The raw status of JPEG\_DCT\_DONE\_INT. (R/WTC/SS)

**JPEG\_BS\_LAST\_BLOCK\_EOF\_INT\_RAW** The raw status of JPEG\_BS\_LAST\_BLOCK\_EOF\_INT.  
(R/WTC/SS)

**JPEG\_SCAN\_CHECK\_NONE\_ERR\_INT\_RAW** The raw status of JPEG\_SCAN\_CHECK\_NONE\_ERR\_INT.  
(R/WTC/SS)

**JPEG\_SCAN\_CHECK\_POS\_ERR\_INT\_RAW** The raw status of JPEG\_SCAN\_CHECK\_POS\_ERR\_INT.  
(R/WTC/SS)

**JPEG\_UXP\_DET\_INT\_RAW** The raw status of JPEG\_UXP\_DET\_INT. (R/WTC/SS)

**JPEG\_EN\_FRAME\_EOF\_ERR\_INT\_RAW** The raw status of JPEG\_EN\_FRAME\_EOF\_ERR\_INT.  
(R/WTC/SS)

**JPEG\_EN\_FRAME\_EOF\_LACK\_INT\_RAW** The raw status of JPEG\_EN\_FRAME\_EOF\_LACK\_INT.  
(R/WTC/SS)

**JPEG\_DE\_FRAME\_EOF\_ERR\_INT\_RAW** The raw status of JPEG\_DE\_FRAME\_EOF\_ERR\_INT.  
(R/WTC/SS)

**JPEG\_DE\_FRAME\_EOF\_LACK\_INT\_RAW** The raw status of JPEG\_DE\_FRAME\_EOF\_LACK\_INT.  
(R/WTC/SS)

**JPEG\_SOS\_UNMATCH\_ERR\_INT\_RAW** The raw status of JPEG\_SOS\_UNMATCH\_ERR\_INT.  
(R/WTC/SS)

**JPEG\_MARKER\_ERR\_FST\_SCAN\_INT\_RAW** The raw status of JPEG\_MARKER\_ERR\_FST\_SCAN\_INT.  
(R/WTC/SS)

**JPEG\_MARKER\_ERR\_OTHER\_SCAN\_INT\_RAW** The raw status of JPEG\_MARKER\_ERR\_OTHER\_SCAN\_INT.  
(R/WTC/SS)

**JPEG\_UNDET\_INT\_RAW** The raw status of JPEG\_UNDET\_INT. (R/WTC/SS)

**JPEG\_DECODE\_TIMEOUT\_INT\_RAW** The raw status of JPEG\_DECODE\_TIMEOUT\_INT. (R/WTC/SS)

### Register 35.16. JPEG\_INT\_ENA\_REG (0x003C)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|-------|
| <div>(reserved)</div> <div>JPEG_DECODE_TIMEOUT_INT_ENA</div> <div>JPEG_UNDET_INT_ENA</div> <div>JPEG_MARKER_ERR_OTHER_SCAN_INT_ENA</div> <div>JPEG_MARKER_ERR_FST_SCAN_INT_ENA</div> <div>JPEG_SOS_UNMATCH_ERR_INT_ENA</div> <div>JPEG_DE_FRAME_EOF_LACK_INT_ENA</div> <div>JPEG_DE_FRAME_EOF_ERR_INT_ENA</div> <div>JPEG_EN_FRAME_EOF_LACK_INT_ENA</div> <div>JPEG_EN_FRAME_EOF_ERR_INT_ENA</div> <div>JPEG_LYP_DET_INT_ENA</div> <div>JPEG_SCAN_CHECK_POS_ERR_INT_ENA</div> <div>JPEG_BS_LAST_BLOCK_NONE_ERR_INT_ENA</div> <div>JPEG_DCT_DONE_INT_ENA</div> <div>JPEG_SR_COLOR_MODE_ERR_INT_ENA</div> <div>JPEG_OUT_EOF_INT_ENA</div> <div>JPEG_RST_CHECK_POS_ERR_INT_ENA</div> <div>JPEG_RST_CHECK_NONE_ERR_INT_ENA</div> <div>JPEG_C_DQT_ID_ERR_INT_ENA</div> <div>JPEG_C_DHT_AC_ID_ERR_INT_ENA</div> <div>JPEG_OID_ERR_INT_ENA</div> <div>JPEG_RLE_PARALLEL_ERR_INT_ENA</div> <div>JPEG_DONE_INT_ENA</div> |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |       |
| 31                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |       |
| 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | Reset |

**JPEG DONE INT ENA** Write 1 to enable JPEG DONE INT ENA. (R/W)

**JPEG RLE PARALLEL ERR INT ENA** Write 1 to enable JPEG RLE PARALLEL ERR INT. (R/W)

**JPEG CID ERR INT ENA** Write 1 to enable JPEG CID ERR INT. (R/W)

**JPEG\_C\_DHT\_DC\_ID\_ERR\_INT\_ENA** Write 1 to enable JPEG\_C\_DHT\_DC\_ID\_ERR\_INT. (R/W)

**JPEG C DHT AC ID ERR INT ENA** Write 1 to enable JPEG C DHT AC ID ERR INT. (R/W)

**JPEG\_C\_DQT\_ID\_ERR\_INT\_ENA** Write 1 to enable JPEG\_C\_DQT\_ID\_ERR\_INT. (R/W)

**JPEG\_RST\_UXP\_ERR\_INT\_ENA** Write 1 to enable JPEG\_RST\_UXP\_ERR\_INT. (R/W)

**JPEG\_RST\_CHECK\_NONE\_ERR\_INT\_ENA** Write 1 to enable JPEG\_RST\_CHECK\_NONE\_ERR\_INT.  
(R/W)

**JPEG\_RST\_CHECK\_POS\_ERR\_INT\_ENA** Write 1 to enable JPEG\_RST\_CHECK\_POS\_ERR\_INT. (R/W)

**JPEG\_OUT\_EOF\_INT\_ENA** Write 1 to enable JPEG\_OUT\_EOF\_INT. (R/W)

**JPEG\_SR\_COLOR\_MODE\_ERR\_INT\_ENA** Write 1 to enable JPEG\_SR\_COLOR\_MODE\_ERR\_INT.  
(R/W)

**JPEG DCT DONE INT ENA** Write 1 to enable JPEG DCT DONE INT. (R/W)

**JPEG\_BS\_LAST\_BLOCK\_EOF\_INT\_ENA** Write 1 to enable JPEG\_BS\_LAST\_BLOCK\_EOF\_INT. (R/W)

**JPEG\_SCAN\_CHECK\_NONE\_ERR\_INT\_ENA** Write 1 to enable JPEG\_SCAN\_CHECK\_NONE\_ERR\_INT.  
(R/W)

**JPEG\_SCAN\_CHECK\_POS\_ERR\_INT\_ENA** Write 1 to enable JPEG\_SCAN\_CHECK\_POS\_ERR\_INT.  
(R/W)

**JPEG UXP DET INT ENA** Write 1 to enable JPEG UXP DET INT. (R/W)

Continued on the next page...

**Register 35.16. JPEG\_INT\_ENA\_REG (0x003C)**

Continued from the previous page...

**JPEG\_EN\_FRAME\_EOF\_ERR\_INT\_ENA** Write 1 to enable JPEG\_EN\_FRAME\_EOF\_ERR\_INT. (R/W)

**JPEG\_EN\_FRAME\_EOF\_LACK\_INT\_ENA** Write 1 to enable JPEG\_EN\_FRAME\_EOF\_LACK\_INT.  
(R/W)

**JPEG\_DE\_FRAME\_EOF\_ERR\_INT\_ENA** Write 1 to enable JPEG\_DE\_FRAME\_EOF\_ERR\_INT. (R/W)

**JPEG\_DE\_FRAME\_EOF\_LACK\_INT\_ENA** Write 1 to enable JPEG\_DE\_FRAME\_EOF\_LACK\_INT.  
(R/W)

**JPEG\_SOS\_UNMATCH\_ERR\_INT\_ENA** Write 1 to enable JPEG\_SOS\_UNMATCH\_ERR\_INT. (R/W)

**JPEG\_MARKER\_ERR\_FST\_SCAN\_INT\_ENA** Write 1 to enable JPEG\_MARKER\_ERR\_FST\_SCAN\_INT.  
(R/W)

**JPEG\_MARKER\_ERR\_OTHER\_SCAN\_INT\_ENA** Write 1 to enable JPEG\_MARKER\_ERR\_OTHER\_SCAN\_INT.  
(R/W)

**JPEG\_UNDET\_INT\_ENA** Write 1 to enable JPEG\_UNDET\_INT. (R/W)

**JPEG\_DECODE\_TIMEOUT\_INT\_ENA** Write 1 to enable JPEG\_DECODE\_TIMEOUT\_INT. (R/W)

Register 35.17. JPEG\_INT\_ST\_REG (0x0040)

|            |   |   |   |   |   |   |   |                                   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |       |
|------------|---|---|---|---|---|---|---|-----------------------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|-------|
| (reserved) |   |   |   |   |   |   |   | JPEG_DECODE_TIMEOUT_INT_ST        |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |       |
|            |   |   |   |   |   |   |   | JPEG_UNDET_INT_ST                 |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |       |
|            |   |   |   |   |   |   |   | JPEG_MARKER_ERR_OTHER_SCAN_INT_ST |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |       |
|            |   |   |   |   |   |   |   | JPEG_MARKER_ERR_FST_SCAN_INT_ST   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |       |
|            |   |   |   |   |   |   |   | JPEG_SOS_UNMATCH_ERR_INT_ST       |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |       |
|            |   |   |   |   |   |   |   | JPEG_DE_FRAME_EOF_LACK_INT_ST     |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |       |
|            |   |   |   |   |   |   |   | JPEG_DE_FRAME_EOF_LACK_INT_ST     |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |       |
|            |   |   |   |   |   |   |   | JPEG_EN_FRAME_EOF_LACK_INT_ST     |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |       |
|            |   |   |   |   |   |   |   | JPEG_UXP_DET_INT_ST               |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |       |
|            |   |   |   |   |   |   |   | JPEG_SCAN_CHECK_POS_ERR_INT_ST    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |       |
|            |   |   |   |   |   |   |   | JPEG_SCAN_CHECK_NONE_ERR_INT_ST   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |       |
|            |   |   |   |   |   |   |   | JPEG_BS_LAST_BLOCK_EOF_INT_ST     |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |       |
|            |   |   |   |   |   |   |   | JPEG_SR_COLOR_MODE_ERR_INT_ST     |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |       |
|            |   |   |   |   |   |   |   | JPEG_OUT_EOF_INT_ST               |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |       |
|            |   |   |   |   |   |   |   | JPEG_RST_CHECK_POS_ERR_INT_ST     |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |       |
|            |   |   |   |   |   |   |   | JPEG_RST_CHECK_NONE_ERR_INT_ST    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |       |
|            |   |   |   |   |   |   |   | JPEG_C_DQT_ID_ERR_INT_ST          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |       |
|            |   |   |   |   |   |   |   | JPEG_C_DHT_AC_ID_ERR_INT_ST       |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |       |
|            |   |   |   |   |   |   |   | JPEG_C_DHT_DC_ID_ERR_INT_ST       |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |       |
|            |   |   |   |   |   |   |   | JPEG_RLE_PARALLEL_ERR_INT_ST      |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |       |
|            |   |   |   |   |   |   |   | JPEG_DONE_INT_ST                  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |       |
|            |   |   |   |   |   |   |   |                                   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |       |
| 31         |   |   |   |   |   |   |   | 25                                | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 | Reset |
| 0          | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0                                 | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |       |

**JPEG\_DONE\_INT\_ST** The masked interrupt status of JPEG\_DONE\_INT. (RO)

**JPEG\_RLE\_PARALLEL\_ERR\_INT\_ST** The masked interrupt status of JPEG\_RLE\_PARALLEL\_ERR\_INT. (RO)

**JPEG\_CID\_ERR\_INT\_ST** The masked interrupt status of JPEG\_CID\_ERR\_INT. (RO)

**JPEG\_C\_DHT\_DC\_ID\_ERR\_INT\_ST** The masked interrupt status of JPEG\_C\_DHT\_DC\_ID\_ERR\_INT. (RO)

**JPEG\_C\_DHT\_AC\_ID\_ERR\_INT\_ST** The masked interrupt status of JPEG\_C\_DHT\_AC\_ID\_ERR\_INT. (RO)

**JPEG\_C\_DQT\_ID\_ERR\_INT\_ST** The masked interrupt status of JPEG\_C\_DQT\_ID\_ERR\_INT. (RO)

**JPEG\_RST\_UXP\_ERR\_INT\_ST** The masked interrupt status of JPEG\_RST\_UXP\_ERR\_INT. (RO)

**JPEG\_RST\_CHECK\_NONE\_ERR\_INT\_ST** The masked interrupt status of JPEG\_RST\_CHECK\_NONE\_ERR\_INT. (RO)

**JPEG\_RST\_CHECK\_POS\_ERR\_INT\_ST** The masked interrupt status of JPEG\_RST\_CHECK\_POS\_ERR\_INT. (RO)

**JPEG\_OUT\_EOF\_INT\_ST** The masked interrupt status of JPEG\_OUT\_EOF\_INT. (RO)

**JPEG\_SR\_COLOR\_MODE\_ERR\_INT\_ST** The masked interrupt status of JPEG\_SR\_COLOR\_MODE\_ERR\_INT. (RO)

**JPEG\_DCT\_DONE\_INT\_ST** The masked interrupt status of JPEG\_DCT\_DONE\_INT. (RO)

**JPEG\_BS\_LAST\_BLOCK\_EOF\_INT\_ST** The masked interrupt status of JPEG\_BS\_LAST\_BLOCK\_EOF\_INT. (RO)

**JPEG\_SCAN\_CHECK\_NONE\_ERR\_INT\_ST** The masked interrupt status of JPEG\_SCAN\_CHECK\_NONE\_ERR\_INT. (RO)

**JPEG\_SCAN\_CHECK\_POS\_ERR\_INT\_ST** The masked interrupt status of JPEG\_SCAN\_CHECK\_POS\_ERR\_INT. (RO)

**JPEG\_UXP\_DET\_INT\_ST** The masked interrupt status of JPEG\_UXP\_DET\_INT. (RO)

Continued on the next page...

**Register 35.17. JPEG\_INT\_ST\_REG (0x0040)**

Continued from the previous page...

|                                          |                                                                     |
|------------------------------------------|---------------------------------------------------------------------|
| <b>JPEG_EN_FRAME_EOF_ERR_INT_ST</b>      | The masked interrupt status of JPEG_EN_FRAME_EOF_ERR_INT. (RO)      |
| <b>JPEG_EN_FRAME_EOF_LACK_INT_ST</b>     | The masked interrupt status of JPEG_EN_FRAME_EOF_LACK_INT. (RO)     |
| <b>JPEG_DE_FRAME_EOF_ERR_INT_ST</b>      | The masked interrupt status of JPEG_DE_FRAME_EOF_ERR_INT. (RO)      |
| <b>JPEG_DE_FRAME_EOF_LACK_INT_ST</b>     | The masked interrupt status of JPEG_DE_FRAME_EOF_LACK_INT. (RO)     |
| <b>JPEG_SOS_UNMATCH_ERR_INT_ST</b>       | The masked interrupt status of JPEG_SOS_UNMATCH_ERR_INT. (RO)       |
| <b>JPEG_MARKER_ERR_FST_SCAN_INT_ST</b>   | The masked interrupt status of JPEG_MARKER_ERR_FST_SCAN_INT. (RO)   |
| <b>JPEG_MARKER_ERR_OTHER_SCAN_INT_ST</b> | The masked interrupt status of JPEG_MARKER_ERR_OTHER_SCAN_INT. (RO) |
| <b>JPEG_UNDET_INT_ST</b>                 | The masked interrupt status of JPEG_UNDET_INT. (RO)                 |
| <b>JPEG_DECODE_TIMEOUT_INT_ST</b>        | The masked interrupt status of JPEG_DECODE_TIMEOUT_INT. (RO)        |

Register 35.18. JPEG\_INT\_CLR\_REG (0x0044)

|            |   |   |   |   |   |   |                             |    |                    |    |                                    |    |                                  |    |                              |    |                                |    |                                |    |                      |    |                                 |   |                                |   |                                |   |                                |   |                           |   |                              |   |                              |   |                               |   |                   |       |
|------------|---|---|---|---|---|---|-----------------------------|----|--------------------|----|------------------------------------|----|----------------------------------|----|------------------------------|----|--------------------------------|----|--------------------------------|----|----------------------|----|---------------------------------|---|--------------------------------|---|--------------------------------|---|--------------------------------|---|---------------------------|---|------------------------------|---|------------------------------|---|-------------------------------|---|-------------------|-------|
| (reserved) |   |   |   |   |   |   | JPEG_DECODE_TIMEOUT_INT_CLR |    | JPEG_UNDET_INT_CLR |    | JPEG_MARKER_ERR_OTHER_SCAN_INT_CLR |    | JPEG_MARKER_ERR_FST_SCAN_INT_CLR |    | JPEG_SOS_UNMATCH_ERR_INT_CLR |    | JPEG_DE_FRAME_EOF_LACK_INT_CLR |    | JPEG_EN_FRAME_EOF_LACK_INT_CLR |    | JPEG_UXP_DET_INT_CLR |    | JPEG_SCAN_CHECK_POS_ERR_INT_CLR |   | JPEG_BS_LAST_BLOCK_EOF_INT_CLR |   | JPEG_SR_COLOR_MODE_ERR_INT_CLR |   | JPEG_RST_CHECK_POS_ERR_INT_CLR |   | JPEG_C_DQT_ID_ERR_INT_CLR |   | JPEG_C_DHT_AC_ID_ERR_INT_CLR |   | JPEG_C_DHT_DC_ID_ERR_INT_CLR |   | JPEG_RLE_PARALLEL_ERR_INT_CLR |   | JPEG_DONE_INT_CLR |       |
| 31         |   |   |   |   |   |   | 25                          | 24 | 23                 | 22 | 21                                 | 20 | 19                               | 18 | 17                           | 16 | 15                             | 14 | 13                             | 12 | 11                   | 10 | 9                               | 8 | 7                              | 6 | 5                              | 4 | 3                              | 2 | 1                         | 0 |                              |   |                              |   |                               |   |                   |       |
| 0          | 0 | 0 | 0 | 0 | 0 | 0 | 0                           | 0  | 0                  | 0  | 0                                  | 0  | 0                                | 0  | 0                            | 0  | 0                              | 0  | 0                              | 0  | 0                    | 0  | 0                               | 0 | 0                              | 0 | 0                              | 0 | 0                              | 0 | 0                         | 0 | 0                            | 0 | 0                            | 0 | 0                             | 0 | 0                 | Reset |

**JPEG\_DONE\_INT\_CLR** Write 1 to clear JPEG\_DONE\_INT. (WT)

**JPEG\_RLE\_PARALLEL\_ERR\_INT\_CLR** Write 1 to clear JPEG\_RLE\_PARALLEL\_ERR\_INT. (WT)

**JPEG\_CID\_ERR\_INT\_CLR** Write 1 to clear JPEG\_CID\_ERR\_INT. (WT)

**JPEG\_C\_DHT\_DC\_ID\_ERR\_INT\_CLR** Write 1 to clear JPEG\_C\_DHT\_DC\_ID\_ERR\_INT. (WT)

**JPEG\_C\_DHT\_AC\_ID\_ERR\_INT\_CLR** Write 1 to clear JPEG\_C\_DHT\_AC\_ID\_ERR\_INT. (WT)

**JPEG\_C\_DQT\_ID\_ERR\_INT\_CLR** Write 1 to clear JPEG\_C\_DQT\_ID\_ERR\_INT. (WT)

**JPEG\_RST\_UXP\_ERR\_INT\_CLR** Write 1 to clear JPEG\_RST\_UXP\_ERR\_INT. (WT)

**JPEG\_RST\_CHECK\_NONE\_ERR\_INT\_CLR** Write 1 to clear JPEG\_RST\_CHECK\_NONE\_ERR\_INT. (WT)

**JPEG\_RST\_CHECK\_POS\_ERR\_INT\_CLR** Write 1 to clear JPEG\_RST\_CHECK\_POS\_ERR\_INT. (WT)

**JPEG\_OUT\_EOF\_INT\_CLR** Write 1 to clear JPEG\_OUT\_EOF\_INT. (WT)

**JPEG\_SR\_COLOR\_MODE\_ERR\_INT\_CLR** Write 1 to clear JPEG\_SR\_COLOR\_MODE\_ERR\_INT. (WT)

**JPEG\_DCT\_DONE\_INT\_CLR** Write 1 to clear JPEG\_DCT\_DONE\_INT. (WT)

**JPEG\_BS\_LAST\_BLOCK\_EOF\_INT\_CLR** Write 1 to clear JPEG\_BS\_LAST\_BLOCK\_EOF\_INT. (WT)

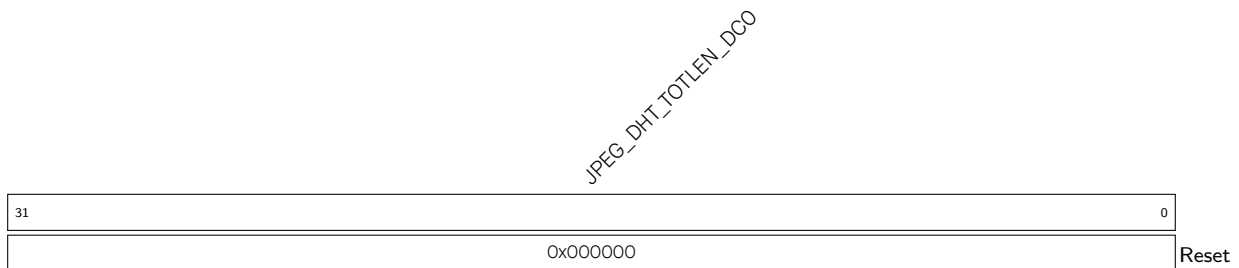
**JPEG\_SCAN\_CHECK\_NONE\_ERR\_INT\_CLR** Write 1 to clear JPEG\_SCAN\_CHECK\_NONE\_ERR\_INT. (WT)

**JPEG\_SCAN\_CHECK\_POS\_ERR\_INT\_CLR** Write 1 to clear JPEG\_SCAN\_CHECK\_POS\_ERR\_INT. (WT)

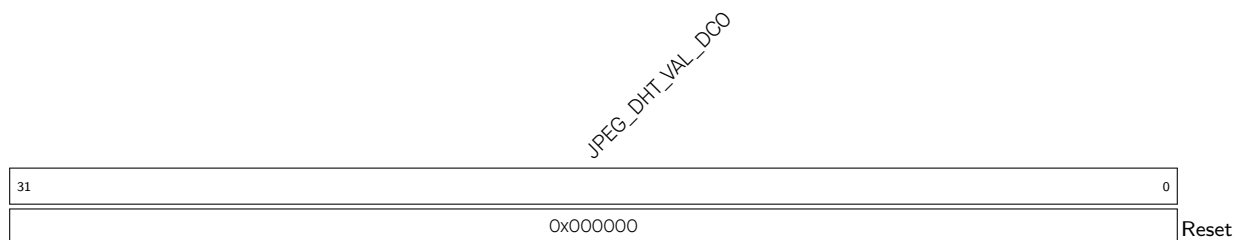
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**Register 35.18. JPEG\_INT\_CLR\_REG (0x0044)**

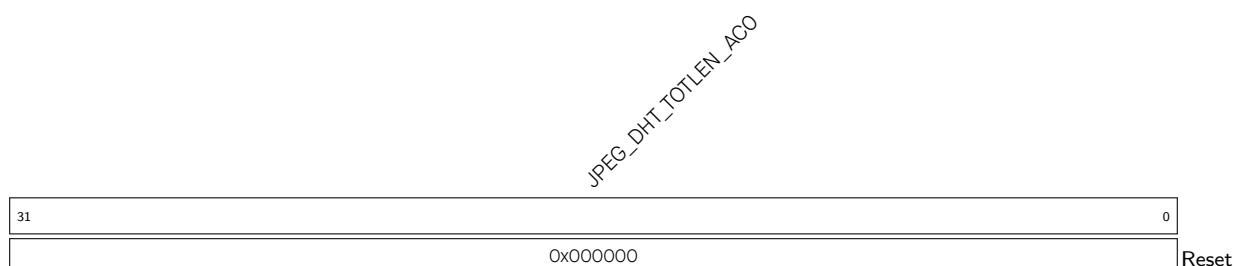
Continued from the previous page...

**JPEG\_UXP\_DET\_INT\_CLR** Write 1 to clear JPEG\_UXP\_DET\_INT. (WT)**JPEG\_EN\_FRAME\_EOF\_ERR\_INT\_CLR** Write 1 to clear JPEG\_EN\_FRAME\_EOF\_ERR\_INT. (WT)**JPEG\_EN\_FRAME\_EOF\_LACK\_INT\_CLR** Write 1 to clear JPEG\_EN\_FRAME\_EOF\_LACK\_INT. (WT)**JPEG\_DE\_FRAME\_EOF\_ERR\_INT\_CLR** Write 1 to clear JPEG\_DE\_FRAME\_EOF\_ERR\_INT. (WT)**JPEG\_DE\_FRAME\_EOF\_LACK\_INT\_CLR** Write 1 to clear JPEG\_DE\_FRAME\_EOF\_LACK\_INT. (WT)**JPEG\_SOS\_UNMATCH\_ERR\_INT\_CLR** Write 1 to clear JPEG\_SOS\_UNMATCH\_ERR\_INT. (WT)**JPEG\_MARKER\_ERR\_FST\_SCAN\_INT\_CLR** Write 1 to clear JPEG\_MARKER\_ERR\_FST\_SCAN\_INT.  
(WT)**JPEG\_MARKER\_ERR\_OTHER\_SCAN\_INT\_CLR** Write 1 to clear JPEG\_MARKER\_ERR\_OTHER\_SCAN\_INT.  
(WT)**JPEG\_UNDET\_INT\_CLR** Write 1 to clear JPEG\_UNDET\_INT. (WT)**JPEG\_DECODE\_TIMEOUT\_INT\_CLR** Write 1 to clear JPEG\_DECODE\_TIMEOUT\_INT. (WT)**Register 35.19. JPEG\_DHT\_TOTLEN\_DCO\_REG (0x0058)****JPEG\_DHT\_TOTLEN\_DCO** Configures the number of codeword with a codeword length of 1 to 16 bits for DCO Huffman table in FIFO mode. (HRO)

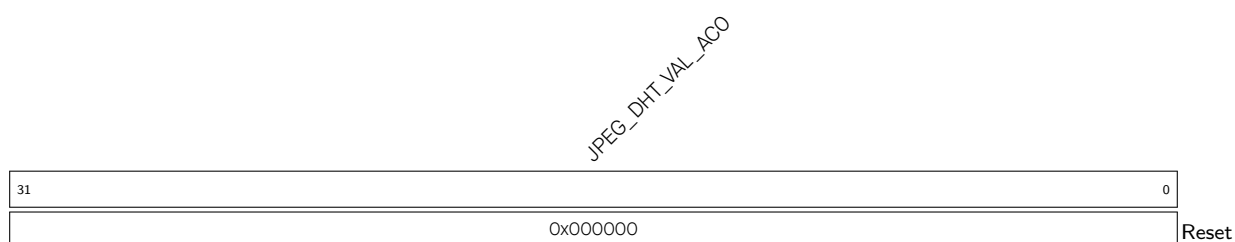


**Register 35.20. JPEG\_DHT\_VAL\_DCO\_REG (0x005C)**

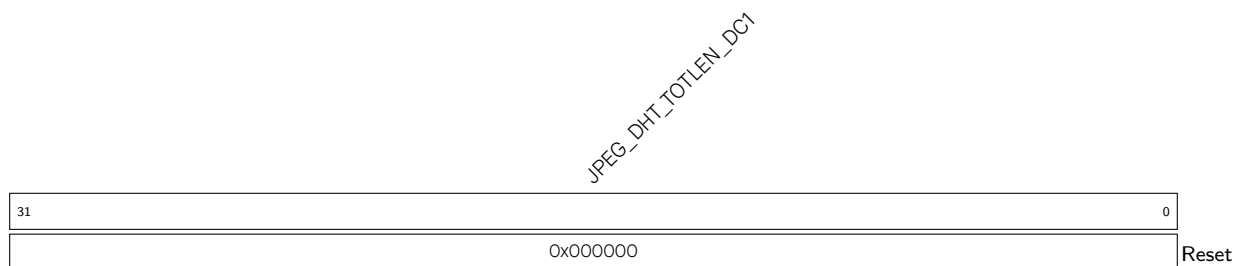
**JPEG\_DHT\_VAL\_DCO** Configures the symbols corresponding to the decoded codeword for DCO Huffman table in FIFO mode. (HRO)

**Register 35.21. JPEG\_DHT\_TOTLEN\_ACO\_REG (0x0060)**

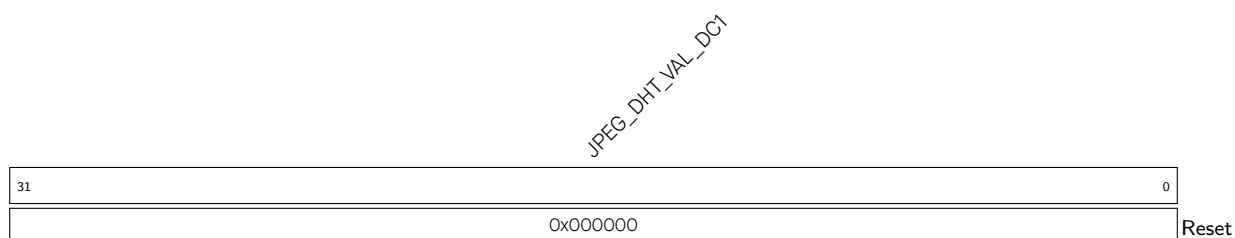
**JPEG\_DHT\_TOTLEN\_ACO** Configures the number of codeword with a codeword length of 1 to 16 bits for ACO Huffman table in FIFO mode. (HRO)

**Register 35.22. JPEG\_DHT\_VAL\_ACO\_REG (0x0064)**

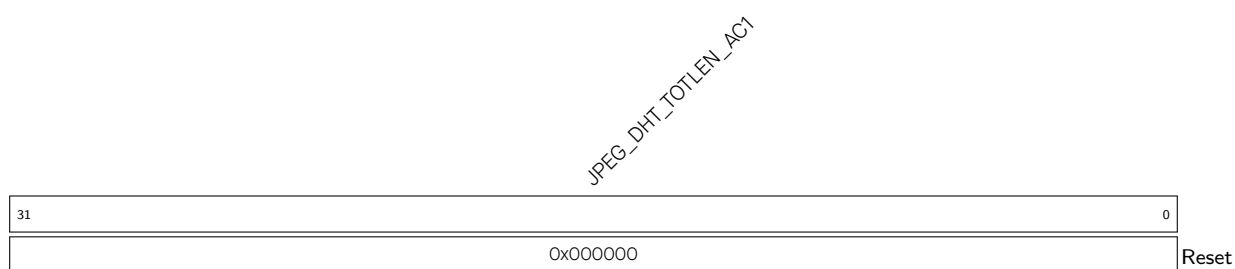
**JPEG\_DHT\_VAL\_ACO** Configures the symbols corresponding to the decoded codeword for ACO Huffman table in FIFO mode. (HRO)

**Register 35.23. JPEG\_DHT\_TOTLEN\_DC1\_REG (0x0068)**

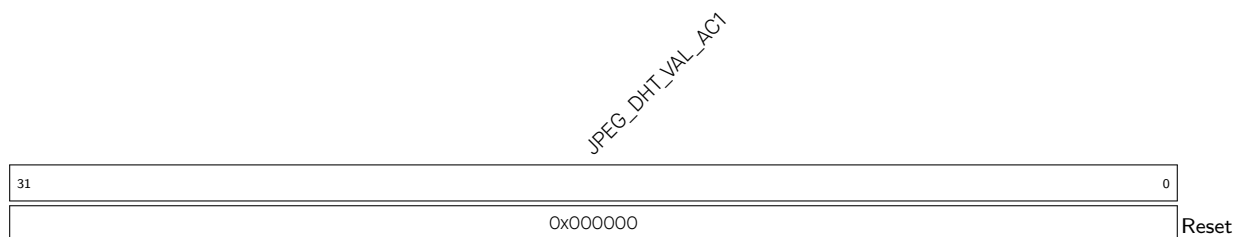
**JPEG\_DHT\_TOTLEN\_DC1** Configures the number of codeword with a codeword length of 1 to 16 bits for DC1 Huffman table in FIFO mode. (HRO)

**Register 35.24. JPEG\_DHT\_VAL\_DC1\_REG (0x006C)**

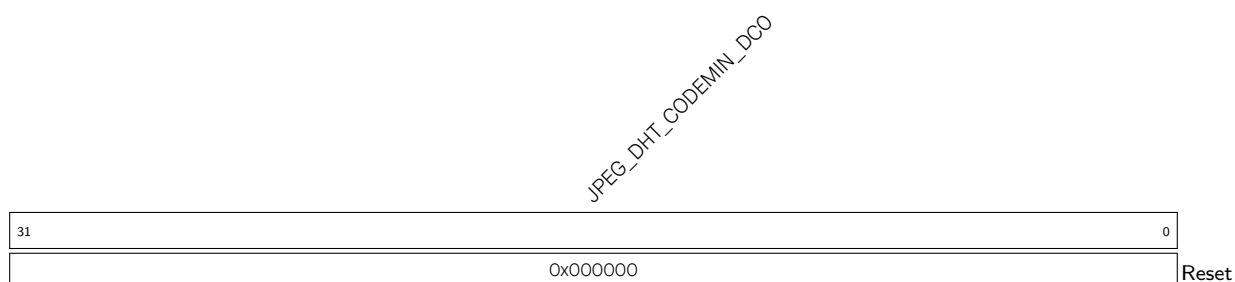
**JPEG\_DHT\_VAL\_DC1** Configures the symbols corresponding to the decoded codeword for DC1 Huffman table in FIFO mode. (HRO)

**Register 35.25. JPEG\_DHT\_TOTLEN\_AC1\_REG (0x0070)**

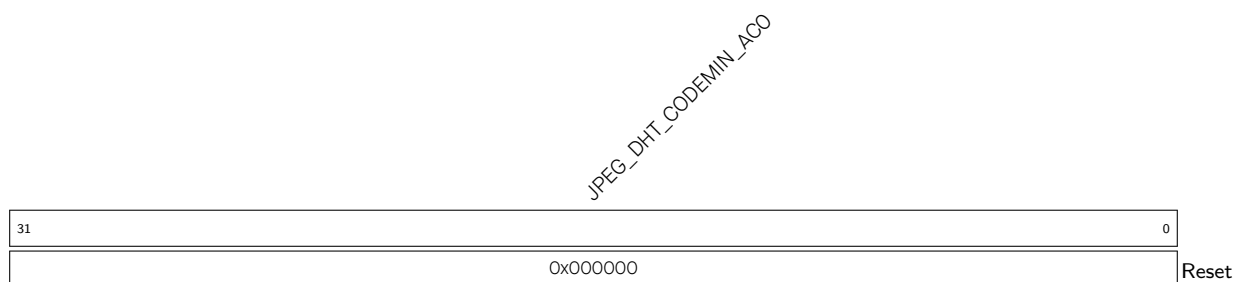
**JPEG\_DHT\_TOTLEN\_AC1** Configures the number of codeword with a codeword length of 1 to 16 bits for AC1 Huffman table in FIFO mode. (HRO)

**Register 35.26. JPEG\_DHT\_VAL\_AC1\_REG (0x0074)**

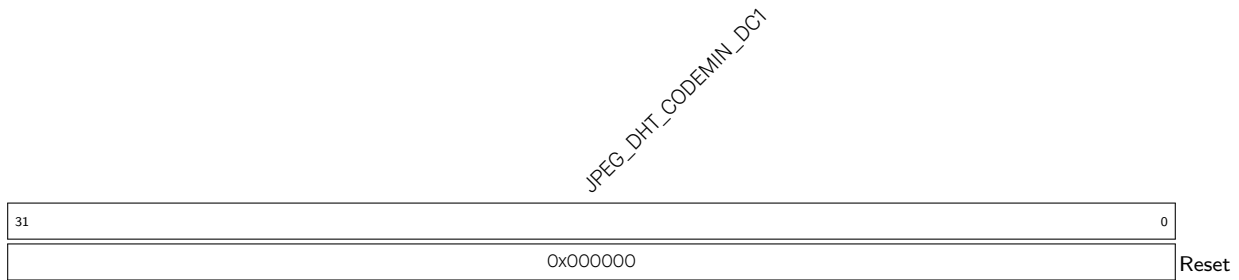
**JPEG\_DHT\_VAL\_AC1** Configures the symbols corresponding to the decoded codeword for AC1 Huffman table in FIFO mode. (HRO)

**Register 35.27. JPEG\_DHT\_CODEMIN\_DCO\_REG (0x0078)**

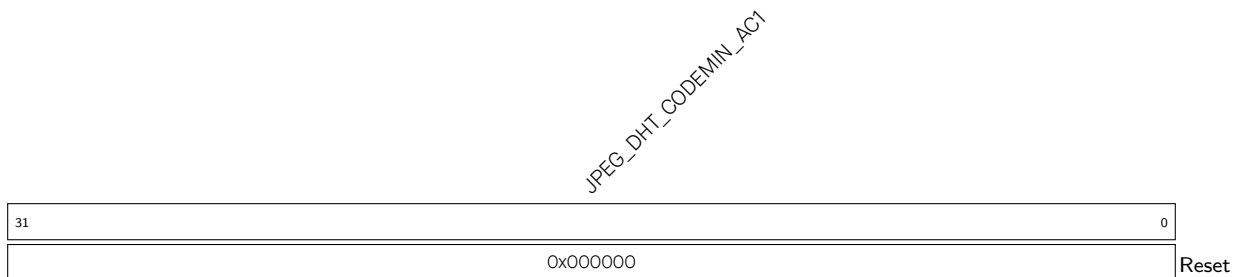
**JPEG\_DHT\_CODEMIN\_DCO** Configures the minimum codeword with a codeword length of 1 to 16 bits for DCO Huffman table in FIFO mode. The codeword should be left shifted to the MSB position of a 16-bit word. (HRO)

**Register 35.28. JPEG\_DHT\_CODEMIN\_ACO\_REG (0x007C)**

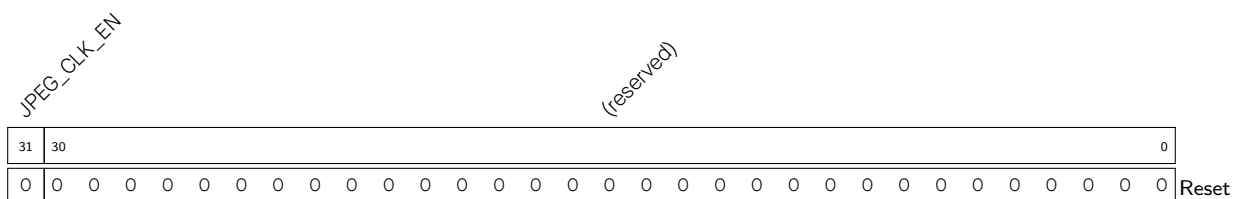
**JPEG\_DHT\_CODEMIN\_ACO** Configures the minimum codeword with a codeword length of 1 to 16 bits for ACO Huffman table in FIFO mode. The codeword should be left shifted to the MSB position of a 16-bit word. (HRO)

**Register 35.29. JPEG\_DHT\_CODEMIN\_DC1\_REG (0x0080)**

**JPEG\_DHT\_CODEMIN\_DC1** Configures the minimum codeword with a codeword length of 1 to 16 bits for DC1 Huffman table in FIFO mode. The codeword should be left shifted to the MSB position of a 16-bit word. (HRO)

**Register 35.30. JPEG\_DHT\_CODEMIN\_AC1\_REG (0x0084)**

**JPEG\_DHT\_CODEMIN\_AC1** Configures the minimum codeword with a codeword length of 1 to 16 bits for AC1 Huffman table in FIFO mode. The codeword should be left shifted to the MSB position of a 16-bit word. (HRO)

**Register 35.31. JPEG\_SYS\_REG (0x00F8)**

**JPEG\_CLK\_EN** Configures whether or not to force open register clock gate.

0: Open the clock gate only when the application writes registers

1: Force open the clock gate for register

(R/W)

Register 35.32. JPEG\_VERSION\_REG (0x00FC)

|            |    |    |   |               |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |
|------------|----|----|---|---------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|-------|
| (reserved) |    |    |   | JPEG_JPEG_VER |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |
| 31         | 28 | 27 | 0 |               |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |
| 0          | 0  | 0  | 0 | 0x2111190     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Reset |

JPEG\_JPEG\_VER   Version control. (R/W)